The Human Well-Being Outcomes of Tree Plantations in Sub-Saharan Africa: A Reassessment of Evidence using Longitudinal Subnational-Year Data

Abstract:

Several globally driven endeavors endorse tree plantations as an intervention for resource restoration, with assumptions of socio-economic outcomes as co-benefits. Yet, evidence of such co-benefits remains questionable and unclear. Focusing on sub-Saharan Africa (SSA), we empirically reassess the effectiveness of tree planting and species diversity in tree plantations on Gross National Income (GNI) per capita, Human Development Index (HDI), and International Wealth Index (IWI). We use multi-way fixed effects instrumental variables on subnational-year panel data on tree plantations in the first-level administrative subdivisions of 12 SSA countries for the years 1992-2016. We find that dedicating more land to tree plantation over a three- to five-year period raises IWI by more than a third and GNI per capita by a substantial annual amount of 205 to 395 USD. We also find positive effects of tree species diversification in plantations (tree crops and planted forests) on GNI per capita and IWI. However, there is no evidence of HDI increments due to tree planting programs. Our findings remain consistent across different specifications, and a mediation analysis stresses that the impacts of tree plantations are led by men's employment opportunities outside the agricultural sector. We conclude that restoration efforts heavily based on tree planting activities can improve economic well-being outcomes, with heterogeneity across locations and genders, when tree species planted are diversified and successfully grown over time.

Keywords: Tree plantations, Economic well-being, Human development, Africa, Panel data.

Highlights

- Tree planting improves International Wealth Index but not Human Development Index in sub-Saharan Africa.
- Tree planting raises Gross National Income per capita only in Western and Central Africa.
- Men's off-farm employment opportunities are one primary mediator to increases in GNI per capita.
- Diversification of tree species raises Gross National Income per capita and International Wealth Index.

1 Introduction

The recent wave to restore degraded forested landscapes worldwide for multiple socio-ecological benefits has bolstered a redeployment of tree plantations as an approach to reforestation efforts. In Africa, thirty-three out of fifty-four countries (61%) are set to restore their degraded forested landscapes (Djenontin et al., 2021; Fagan et al., 2020) to counter the alarming loss rate of forests that is estimated at 3.94 million ha/year from 2010-2020 and regain the 65% of productive land that is degraded (Mansourian and Berrahmouni, 2021). The rationale underlying increasing and maintaining tree and forest cover lies in the indisputable vital role that trees and forests play by providing multiple ecosystem services and functions such as carbon sequestration, water and climate regulation, habitat, and biodiversity support, as well as food and energy supply, and cultural services that contribute to livelihoods and human well-being (Miller et al., 2022, 2021; Razafindratsima et al., 2021; Reed et al., 2017). It is also established that poorer and vulnerable communities rely on various forest and tree resources to make a living, earning income and providing subsistence (Oyono et al., 2005; Mamo et al., 2007; Mekonnen et al., 2013 & 2014; Malleson et al., 2014; Jasaw et al., 2015; Mutandwa and Kanyarukiga, 2016; Chagumaira et al., 2016; Endamana et al., 2019; Turner-Skoff and Cavender, 2019), with estimates of 1.6 billion forest-proximate people globally and 1.2 billion nature-dependent people in the tropics (Newton et al., 2020; Fedele et al., 2021). As such, efforts to restore forested landscapes are critical for addressing deforestation, land degradation, and climate change challenges and for providing economic and social benefits for rural communities and small-scale farmers that rely heavily on these resources for livelihoods.

In this context, tree planting appears as a preferred and supposedly easy-fix intervention choice to achieve such forest restoration goals and expected benefits (Hawes, 2018; Brancalion and Holl, 2020; Erbaugh et al., 2020), drawing increasing attention to their impacts on socio-economic development alongside environmental sustainability. Tree planting and plantations are not new tools or instruments in environmental governance (Payn et al., 2015; Korhonen et al., 2020). In many African countries, tree planting campaigns are a colonial legacy that has endured and continued to prevail in postcolonial environmental management and forestry (Davis and Robbins, 2018). In some cases, such simplistic instrumental solutions originated from misconceptions and misreading of African landscapes (Fairhead et al., 1996), while in other instances, they were based on not-well-discerned understandings of complex and intertwined drivers of environmental degradation (Bassett and Crummey, 2003; Turner et al., 2023). It is worth noting that the extent and effectiveness of tree planting and af/reforestation programs can vary widely between countries and regions, and some countries may face challenges in implementing and maintaining these initiatives due to factors such as resource and institutional constraints, political instability, or lack of public support (Etongo et al., 2015; Adams et al., 2016; Turner et al., 2023). Strong voices against persisting simplistic, but powerful, tree planting discourses are mounting (Holl and Brancalion, 2020), with calls for nuances and to move toward "tree growing" (Duguma et al., 2020) and assisted natural regeneration methods (Uriarte and Chazdon, 2016; Reij and Garity, 2016; Murugan and Israel, 2017; Chazdon et al., 2020). Yet, many actions continue to foster tree planting to establish plantations as silver bullet solutions (Bastin et al., 2019), enabling powerful initiatives such as the Trillion Trees initiative by the World Economic Forum's (https://www.1t.org/) (Lippke et al., 2021). Such advanced tree planting and plantation interventions align with the global interests in the decarbonization potential of forest restoration (Busch et al., 2019; Bernal et al., 2018; Griscom et al., 2017) and incentives for carbon trading despite their limitations and pitfalls (Kansanga and Luginaah, 2019; Coleman et al., 2021; Fleischman et al., 2020) and sometimes colonialism tone (Bumpus and Liverman, 2010; Lyons and Westoby, 2014). Support for tree planting efforts also resonates with their articulated significance in restoring degraded landscapes and in contributing to forest-based bioeconomy, notably through their potential to enhance the socio-economic needs and livelihoods of rural communities (Blay, 2012; Turner-Skoff and Cavender, 2019; Kimambo et al., 2020; Rosa and Martius, 2021).

Still, the extent to which the reinforced and persisting rush to tree planting and plantations is the right policy intervention lever to realize the undeniable social benefits that tree and forest resources offer remains questionable and unclear (Gerber, 2011; Rana and Miller, 2021). While Rana and Miller (2021) offer insights into factors that can help to understand the impacts of tree planting programs, actual

quantified evidence remains to be substantiated. Recent systematic reviews and meta-analyses point to the limited evidence or, at best, mixed evidence on the long-term socio-economic and human well-being impacts of tree plantations, calling for more evidence-based knowledge in different contexts through empirical assessments (Adams et al., 2016; Sloan et al., 2019; Malkamäki et al., 2018; Hajjar et al., 2021a). Several challenges affect evidence-making on the human well-being impacts of tree planting activities and tree plantations to support effective decisions. Focusing on methodological shortcomings, a significant improvement in using temporal trends and multi-scale patterns for better evidence-making is needed. Using longitudinal data of households or individuals over time to bring insights into well-being impacts at different scales/units of observation is needed (Jagger et al., 2022). Expanding research on the long-term could help with comparisons over time, thereby allowing for a stronger analysis of human well-being connectivity with tree-based interventions and mediating factors (Oldekop et al., 2021). Another challenge is the narrowness or lack of diversity in geographic coverage. Nearly half of common forest-poverty studies come mainly from just Bangladesh, Brazil, China, India, and Nepal, with poor global distribution reflected in academic research. In Africa, in particular, evidence on livelihood outcomes in West and Central Africa is very limited (Hajjar et al., 2021a). Therefore, additional studies about under/less-covered geographies and their diverse contexts over long-time periods are needed to expand evidence-based knowledge on the potential human well-being effects of tree planting interventions and plantations and shed light on the drivers of those effects.

We reexamine empirically the potential impact of tree plantations on economic and human well-being outcomes in Africa using improved econometric methodologies. We use a three-fold exploration of recent panel data on tree plantation areas and human well-being indicators at the locality level to provide new evidence for policy and management decisions. First, we use subnational-year panel data to evaluate whether tree plantations improve Gross National Income (GNI) per capita, Human Development Index (HDI), and International Wealth Index (IWI), including whether these effects vary by the gender of people whose livelihoods might depend on forests in Western and Central Africa (WCA) and Eastern and Southern Africa (ESA). Second, we investigate the pathways through which tree plantations impact GNI, HDI, and IWI by uncovering the mediating factors of the impacts in WCA and ESA. Finally, we examine whether species diversity in tree plantations enhances well-being across the two SSA regions.

We find that an increase in the area under tree planting improves gross income per capita by a substantial amount of 205 to 395 USD/year. However, this income benefit applies to WCA but not ESA and does not reflect any improvement in the human development index. These results are maintained using increases in the area under tree planting within the last three or five years as the treatment variable. The benefits from tree planting and diversification of tree species in the tree plantations extend to reducing severe child stunting both in WCA and ESA. Moreover, our mediation analyses reveal that improvement in off-farm employment opportunities is a major driving channel linking tree plantations to sub-national income per capita enhancement.

We offer several additions to the empirical literature on tree-planting effects on human well-being. First, by employing a rich and unique panel dataset, we are able to estimate and compare the effects of tree planting for two different SSA regions. Existing studies mostly focused on one country or different countries from the same African region without exploiting within-country variations in tree plantation areas, tree species, and human well-being outcomes. Second, we estimate the impacts of tree planting on well-being by employing a multi–way fixed effects instrumental variable (MWFE-IV) estimation procedure. By doing so, we control for the potential endogeneity of tree planting and the time-invariant and subnational region-invariant unobserved heterogeneity. Most previous studies have employed cross-sectional data. The third contribution relates to the ongoing debate on the effectiveness of tree planting in improving human well-being. Our study contributes to estimating and confirming further the nature of human well-being benefits of tree/forest plantations and the possible mediating pathways, as well as the trade-offs associated with the promotion of large-scale tree planting as a nature-based intervention lever to achieve resource restoration for socio-ecological well-being. Restoration efforts heavily based on tree planting activities may offer the assumed and claimed well-being co-benefits, notably income through employment, on a relatively medium term. This is however heterogeneous across locations and genders and materializes only when tree species planted are diversified and can endure, i.e., grow. We draw

attention to the need to put in place necessary resources, infrastructures, technical expertise for managing and monitoring tree planting and plantations, and institutional enabling conditions as such conditions can contribute to the realization of socio-economic benefits and accrued well-being effects from the ecosystem services and functions that tree systems offer.

In the remainder of the paper, we outline the methodology employed, including describing the data sources used and the econometric methods applied in Section 2. We present our results in Section 3, emphasizing salient findings, which we discuss in Section 4 before concluding in Section 5.

2 Social impacts of tree plantations and conceptualization of mediation factors

Social impacts of tree planting programs. Previous research informing on the social and human well-being impacts of tree planting activities and tree plantations indicate both short-term negative and positive impacts (Adams et al., 2016; Kuyah et al., 2019; Malkamäki et al., 2018; Hajjar et al., 2021b; Castle et al., 2021; Peros et al., 2022). Observed evidence suggests local-level socioeconomic benefits of tree-based activities such as income, land productivity, household resilience, food security, and access to ecosystem services (Adams et al. 2016; Shyamsundar et al., 2021). For example, agroforestry and tree cash crops can contribute to food security and alleviate the risk of poverty by creating additional income sources from sales of diverse products, thereby contributing toward the well-being of the nearby communities (Razafindratsima et al., 2021). Also, both positive and negative human well-being impacts of restoration interventions using plantations of exotic/commercial tree species are documented (Davis and Robbins, 2018; Malkamäki et al., 2018; Razafindratsima et al., 2021; McElwee and Nghi, 2021). In Chile for instance, participants to such schemes perceived an increase in opportunities for small-scale enterprises after pine plantations. Regardless, the presence of the plantation prevented other possible land uses and, where created, plantation jobs are often temporary, part-time, or both (Malkamäki et al., 2018). In Vietnam, large-scale tree planting programs, fostered as part of restoration efforts, produced both positive and negative social impacts, but overall, the positive social and institutional benefits were little valued by local communities and smallholder farmers (McElwee and Nghi, 2021). In Tanzania, eucalyptus and pine plantations were detrimental to the livelihoods and income generation of the rural poor as these disrupted traditional farming systems that support healthy food production (Mwanukuzi, 2009). Similarly with *Jatropha curcas* tree crops in Zambia had detrimental effects on the livelihoods of small-scale farmers (Kalinda et al., 2015). Conversely, in the framework of the Great Green Wall's tree planting activities in Africa, income-generating activities have positive impacts on surrounding households. These include the sale of non-timber forest products, of seedlings, and of fodder produced through integrated land management as well as the provision of labor for tree-planting based restoration efforts (Duboz et al., 2019; Jones and Rosenstock, 2021). Nonetheless, more recent studies deemed these as fleeting social impacts that are not based on actual empirical assessments (Turner et al., 2023) and while positive health impacts are further speculated about, clear evidence is yet to be established (Duboz et al., 2019). Besides, tree plantations were found to support timber and fuelwood harvest, which reduces household expenses as well as time to gather fuelwood and allow more free time to invest in other endeavors (Razafindratsima et al. 2021). Still, while market-oriented plantations that center timber products can produce profits, the majority are distributed to the wealthy and politicians, contrasting the benefits of planted fruit trees in terms of a significant increase in food consumption that contributed to diversifying household consumption and children's dietary habits (Razafindratsima et al., 2021). The foregoing suggests not only mitigated evidence of the aggregate potential human well-being effects of tree planting and plantations interventions, including less clarity on the mediators of the effects, but also challenges in uncovering these. This study contributes to addressing that by drawing on longitudinal data and improved econometric methodologies.

Conceptualizing mediation factors. Understanding what mediates the human well-being impacts of tree plantations is important to further establish the potential routes through which tree plantations lead to estimated social outcomes. Mediation analysis helps to assess both the direct causal mechanisms or the indirect pathways (via mediators or intermediaries), through which an intervention (here tree/forest plantations) produces an outcome (here human well-being) (Huber, 2019; Frolich and Huber, 2017; Acharya et al., 2016). To support exploration of possible mediators, we conceptualize a theory of change that considers three main pathways, including employment opportunities (both non-agricultural and agricultural), tree species diversification, and environmental factors (Figure 1).

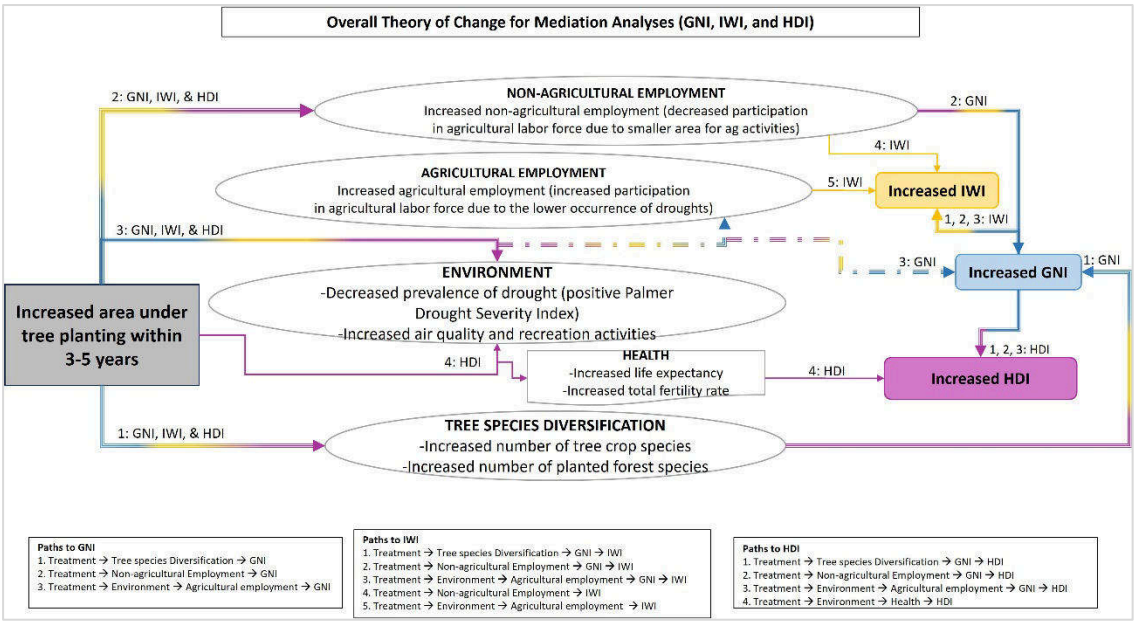


Figure 1: Hypothesized direct and indirect pathways linking tree and forest plantations to human well-being outcomes

First, an increased area under tree plantations can offer additional employment opportunities in rural areas via labor-intensive plantation activities (Wang et al., 2022). For instance, agroforestry systems require more year-round labor than seasonal annual crops (Waldron et al., 2017). However, expansion of tree plantations could displace smallholder farmland and reduce agricultural wage work (Rudel et al., 2009). The net impacts of tree plantations on employment and income are context specific. Increased income from tree plantations enables the rural poor to diversify their livelihoods, thus reducing poverty (Miller et al., 2022). Tree crops have also been linked to decreased wealth inequality in some regions by disproportionately benefiting smallholder farmers (Jin et al., 2021).

Second, tree plantations can enhance human well-being through diversification of the planted species. Diversification supplements households' income sources and enables them to meet basic needs during lean seasons or crop failures (Miller et al., 2022; Sendzimir et al., 2011). For instance, non-timber food products made available from enhanced by tree crop diversity are important for safety net (Adams et al. 2016; Shyamsundar et al., 2021; Razafindratsima et al., 2021; Djenontin and Djoudi, 2015). Moreover, planting diverse tree species mitigates income variance from shocks impacting a single commodity (Lin, 2011).

Third, tree plantations can influence human well-being through environmental factors such as drought reduction and air quality improvement, with incidence on health. Planted forests can increase regional rainfall, reduce temperatures, improve soil health, and alleviate local drought severity (Ellison et al., 2017). Reduced drought incidence enables more consistent agricultural production and income generation, as rural households are less likely to lose crops or livestock to water scarcity. Furthermore, trees filter pollutants from the air and exude oxygen through photosynthesis, resulting in improved local air quality (Nowak et al., 2014). Cleaner air reduces respiratory and cardiovascular diseases, lowering mortality and morbidity, and improving life expectancy and fertility rates (Pope et al., 2009). Further, recreation in green spaces provides mental health benefits that enhance human well-being (Hartig et al., 2014).

Overall, our study aims to substantiate understanding of employment, tree crop diversification, and environmental pathways as conceptual mediators of tree planting effects on human well-being. However, there are likely additional channels that we are unable to investigate empirically due to data limitations. While an exhaustive analysis of all possible pathways goes beyond the scope of this study, we aimed at rigorously investigating the economic and human development impacts that are feasible to measure with available data, notably GNI, HDI, and IWI.

3 Data and methods

1.1 Data

The study data is a panel sample covering 25 years of record (1992 through 2016) of tree plantations (tree crops and planted forests) for the first-level administrative divisions of 12 SSA countries (Benin, Congo Democratic Republic, Cote d'Ivoire, Ghana, Guinea, Rwanda, and Togo in Western and Central Africa; Kenya, Lesotho, Mozambique, Namibia, and Zimbabwe in Eastern and Southern Africa) as illustrated in Figure 2. To depart from previous literature on the social impact of tree plantations in SSA using panel data estimations, we use country-region-year observations, where a region is a sub-national division (administrative level 1) of a country. Data analysis at the country-region-year level to measure the effects of tree plantations in the specific regions where the local people live is more appropriate than the one at the country-year level, which may suffer from aggregation biases: tree plantations might have a well-being impact in some divisions of a country, while the impact will be missed out when measuring the effects of tree plantations at the national level; likewise, tree plantations might have no well-being impact in any division, but some impact might be observed when aggregating all sub-national tree plantations at the national level.

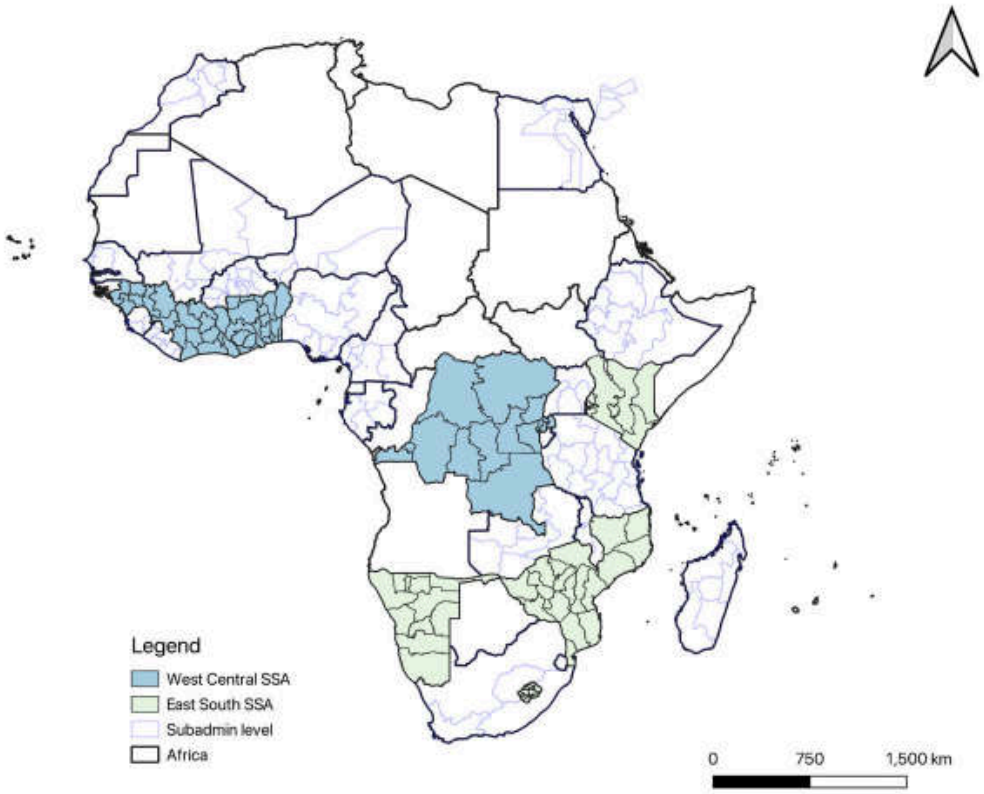


Figure 2: Countries and respective regions included in the study.

Sub-national data analysis is made possible by recent efforts from development and research agencies to release public datasets at the sub-national level for human development indicators,[1] demographic, and health indicators,[2] and global tree planting data (Du et al., 2022). We source annual sub-national level data for Gross National Income (GNI) per capita in USD (2011 PPP[3]), Human Development Index (HDI), employment, life expectancy, and education for the regions and by gender, the International Wealth Index (IWI), population, and total fertility rate from the Global Data Lab (https://globaldatalab.org/areadata). As the overarching measure of human well-being, HDI accounts for both monetary and non-monetary human development. We complement it with two other human well-being outcomes: the GNI which serves as a money-metric standard, income, and the IWI which offers an alternative metric that is correlated to HDI, GNI, life expectancy, and other poverty-based

[1] https://globaldatalab.org/areadata and https://hdr.undp.org/data-center/documentation-and-downloads
[2] https://globaldatalab.org/areadata
[3] Purchasing Power Parity

indicators (Smits and Steendijk, 2015). IWI is based on data from over 2.1 million households in 97 low and middle-income countries, and includes assets like a TV, fridge, phone, car, a house with piped drinking water, electricity, a flush toilet, good quality floor material, and three or more rooms. Households with an IWI value of 100 have reached a standard of material well-being that is considered reasonable even in high-income countries, while households with an IWI value of 0 own none of the included items, have a floor of earth or dung, have no or poor-quality toilet, no electricity, only one room, and water from an unprotected source (Smits and Steendijk, 2015).

We merge the data on the human well-being indicators with the global tree crops and planted forest data (Du et al., 2022). The global tree crops and planted forests data provides us with a uniquely available high-resolution pixel-level spatial data of plantations at the time of our analysis. Indeed, this global plantation extent dataset builds on exiting high-resolution planted trees dataset from the Spatial Database of Planted Trees (SDPT) (Harris et al., 2021) and global palm oil plantations from Descals et al. (2021). Covering planted forests and tree crops at a spatial resolution of 30 meters, in GeoTIFF format, it offers detailed information on tree species and year of planting (Du et al., 2022). The "plantyear" and "species" bands offer precise information on the year of planting and type of tree, respectively, with values ranging from 1981 to 2019 and 1 to 190. Using rasters provided by Du et al (2022) in Google Earth Engine, we extracted the "plantyear" and "species" information for each pixel and subsequently aggregate the data using our shapefile for country-region data. Each pixel in Du et al.'s (2022) dataset represents a 30m area under tree planting, which allows us to obtain sub-national level tree planting area in square kilometers. Overlaying the dataset with primary forest data (Turubanova et al., 2018) helps us to distinguish primary forests from tree plantations/planted forests (See Figure A1 in Appendix).

Lastly, we add to these subnational-year-level data, various country-year-level data used as control variables, instruments, intermediary outcomes and/or mediation factors. These include education, life expectancy, and the share of women in parliament from the UNDP's Human Development Report data center, [4] population, employment in agriculture, age at first birth of women aged 20-50, and total fertility rate data from the Global Data Lab (https://globaldatalab.org/areadata), and climate and environmental factors. We collected Palmer Drought Severity Index (PDSI) data from the TerraClimate dataset. To access this information, we used a subnational shapefile in Google Earth Engine. TerraClimate is a comprehensive monthly climate and water balance dataset covering worldwide terrestrial surfaces. It uses a water balance model that considers reference evapotranspiration, precipitation, temperature, and interpolated plant extractable soil water capacity to produce monthly surface water balance datasets. We further retrieved oil palm suitability information (Strona et al., 2018). The mean oil palm suitability index is a measure of how suitable the land is for oil palm cultivation. The data, originally drawn from the Food and Agriculture Organization of the United Nations Global Agro-Ecological Zones database, is presented in a raster format and ranges from 1 to 8 for high to no suitability pixels, based on rainfed oil palm suitability. The resolution is 5 arc minutes, which is equivalent to approximately 10 km at the equator. We utilized QGIS to calculate the mean suitability at the sub-national shapefile level (See Figure A2 in appendix). The higher the value taken by the suitability index, the lower the oil palm suitability. This index variable is later used to compute instruments for oil palm tree plantations, as described in Section 2.2.2. In the robustness checks of our analyses (Supplemental file, section 1.2), we also conduct estimations that controlled for primary forest characteristics, sourced from the Global Forest Watch database. See Table A1 in appendix for a summary of these different data and their sources.

Table 1 presents the descriptive statistics of tree planting data (Panel A), instrument characteristics (Panel B), GNI per capita, HDI, and IWI data (Panel C), employment in agriculture, health, and environmental characteristics (Panel D), and demographic and institutional characteristics (Panel E).

Table 1: Description of the sample at the subnational-year level

	Obs	Min	Mean	SD	Max
Panel A: Tree planting characteristics					

[4] https://hdr.undp.org/data-center/documentation-and-downloads

Annual area under tree planting (km2)	2700	0	2.075	9.517	178.1
Annual area under tree planting is 0 km2[a]	2700	0	0.551	0.497	1
Annual area under tree planting is between 0 and 1 km2[a]	2700	0	0.317	0.465	1
Annual area under tree planting is between 1 and 10 km2[a]	2700	0	0.0767	0.266	1
Annual area under tree planting is above 10 km2[a]	2700	0	0.0552	0.228	1
Annual area under tree crops (km2)	2700	0	0.872	6.656	158.6
Annual area under planted forest (km2)	2700	0	1.202	6.076	132.9
Increased area under tree planting within the last 3 years (triennial)[a]	2700	0	0.321	0.467	1
Increased area under tree planting within the last 4 years (quadrennial)[a]	2700	0	0.355	0.479	1
Increased area under tree planting within the last 5 years (quinquennial)[a]	2700	0	0.378	0.485	1
Panel B: Instrument characteristics					
Mean palm oil suitability index	2700	2.655	7.146	1.498	8.157
Previous share of other regions with increased tree planting area (triennial)	2700	0	0.331	0.257	0.933
x Mean palm oil suitability index	2700	0	0.0226	0.0180	0.0734
Previous share of other regions with increased tree planting area (quadrennial)	2700	0	0.366	0.263	0.933
x Mean palm oil suitability index	2700	0	0.0250	0.0185	0.0734
Previous share of other regions with increased tree planting area (quinquennial)	2700	0	0.392	0.264	0.933
x Mean palm oil suitability index	2700	0	0.0268	0.0186	0.0734
Panel C: Income and human development characteristics					
GNI per capita in thousands of USD (2011 PPP)	2700	0.360	2.976	3.409	27.08
GNI per capita in thousands of USD, female (2011 PPP)	961	0.479	2.708	2.939	19.64
GNI per capita in thousands of USD, male (2011 PPP)	961	0.562	3.953	4.102	35.40
Human Development Index (value)	2700	0.168	0.443	0.0985	0.726
HDI female	961	0.217	0.455	0.0975	0.737
HDI male	961	0.294	0.512	0.0834	0.717
International Wealth Index (IWI)	2253	0.010	29.59	15.08	77.50
Panel D: Employment, health, and environment characteristics					
Percent of employed women working in agriculture	2253	0	39.83	27.13	99.60
Percent of employed men working in agriculture	2253	0.510	41.10	23.26	89.20
Life expectancy at birth (years)	2700	24.51	53.95	6.215	71.84
Life expectancy at birth, female (years)	961	41.37	58.60	6.149	74.83
Life expectancy at birth, male (years)	961	37.73	55.03	6.128	69.61
Total fertility rate	2253	2.340	4.989	1.270	8.850
Subnational-year mean Palmer Drought Severity Index (PDSI)	2700	-11.04	-0.915	2.588	11.63
Panel E: Demographic, institutional, and primary forest characteristics					
Population size in thousands	2700	47.84	1842.5	1973.0	12926.6
Mean age at first birth of women aged 20-50	2253	17.30	19.59	0.943	22.70
Mean years schooling (years)	2700	0.346	4.162	2.224	10.69
Mean years schooling, female (years)	961	0.137	4.150	2.431	10.16
Mean years schooling, male (years)	961	0.725	5.518	2.264	11.36
Share of seats in parliament, female (percent held by women)	2568	1.235	16.62	11.04	57.55
Log primary forest tree cover extent in ha	1615	0.011	11.04	4.869	18.47
Annual loss in primary forest tree cover extent (thousand ha)	1520	0	27.91	80.47	499.6

Notes: This table shows the summary statistics of the study variables using country-region-year observations for the 12 countries over the 1992-2016 period. Gender-disaggregated characteristics are only available for the 2000-2016 period. Primary forest tree cover extent is available from the year 2000 at the country level and does not include Namibia. Primary forest tree cover extent is the lowest for Lesotho (1.011091 ha in 2000) and the highest for Congo Democratic Republic (above 100 million ha in all years). Annual loss in primary forest tree cover extent is available from the year 2001. HDI: Human Development Index. GNI: Gross National Income. USD: United States Dollars. PPP: Purchasing Power Parity.

[a] Indicates dummy variables

The annual area under tree plantations in the subnational-year sample is 2 square kilometers on average (42% of which is occupied by tree crops and 58% by planted forests). About 55% of the sample have no tree plantations, 39% have up to 10 square kilometers of tree plantations, and 6% have tree plantations of above 10 square kilometers (Panel A). Across countries, the annual area under tree plantations increased at an extensive margin in the past three years for about one-third of the sample (15% in the Eastern and Central Africa sub-sample and 48% in the Western and Central Africa sub-sample), and in the past four years for about 36% of the sample (17% in the Eastern and Central Africa sub-sample and 53% in the Western and Central Africa sub-sample).

Panel B provides the summary statistics of the instruments used in this study, namely the lagged leave-

one-out/spatial instrument for the triennial, quadrennial, and quinquennial treatment variables (i.e., lagged share of other regions that have seen triennial, quadrennial, and quinquennial increase in tree plantation area) and the mean oil palm suitability index interacted with the lagged spatial instrument. These two instruments are further presented in detail in Section 2.2.2. The mean oil palm suitability index ranges from 1 to 8, where a score of 1 indicates that the land is most suitable for oil palm cultivation, and a score of 8 indicates that the land is not suitable for oil palm cultivation. We have also computed the lagged share of other regions that have seen an increase in tree plantation areas over the last three, four, and five years, which has similar mean as the treatment in Panel A, and which we interact with the mean oil palm suitability index. This interaction allows us to capture the tree planting behavior in neighboring regions highly suitable for oil palm cultivation, which is an essential factor in assessing land suitability for oil palm cultivation in each region. By combining oil palm suitability index and the lagged spatial instrument, we can better capture expansions in tree plantation areas influenced by land suitability for oil palm cultivation.

Subnational-year GNI per capita is approximately 3,000 USD (2011 PPP) on average but with a large dispersion across regions (Table 1, Panel C). Women's income is about 32% lower than men's income on average in our sample. Sub-national-year HDI, irrespective of gender, ranges from 0.168 to 0.726 in the years 1992-2016, but HDI varies from 0.217 to 0.737 depending on gender in the years 2000-2016. IWI also has a large range, from nearly zero to 78 (Table 1, Panel C). Figure 3 shows little differences in the distribution of the three outcome variables by categories of the binary treatment variable (triennial and quadrennial) and indicates that initiatives expanding tree plantations rise sharply after the year 2000.

On average, in the sampled regions, about forty percent of women and men are employed in agriculture (Table 1, Panel D). Women expect to live longer than men; they start giving birth in their early twenties and have five children on average (Table 1, Panels D).

On average, the educational attainment of a person (woman or man) in the sampled regions is at the primary level (Table 1, Panel E). The subnational-year population of a region varies from 48 thousand to 13 million inhabitants (Table 1, Panel E).

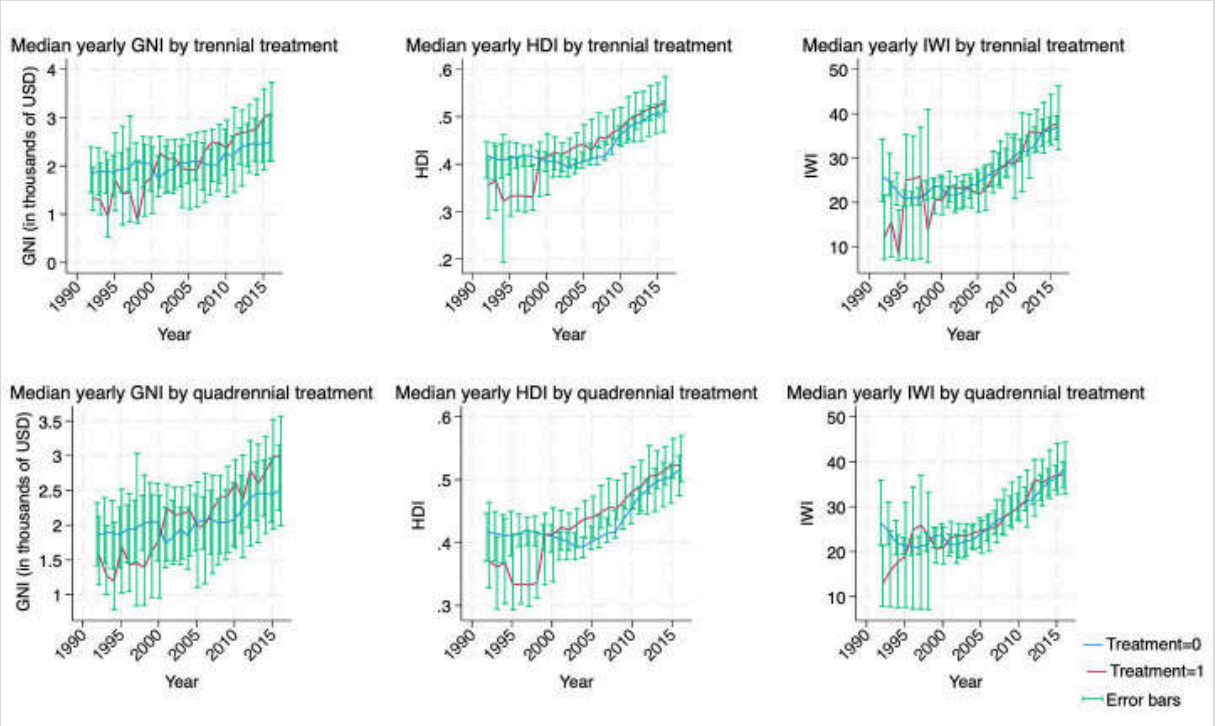
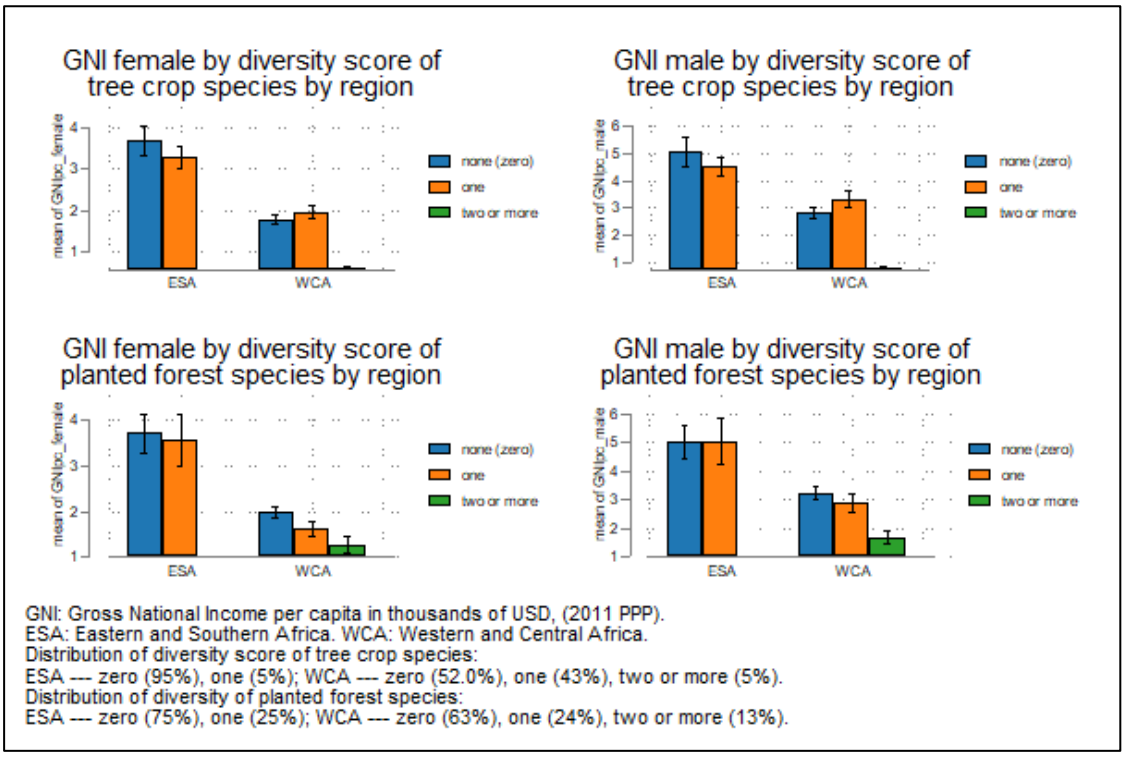


Figure 3: Distribution of GNI, HDI and IWI by treatment variable

We further compute a diversity score for tree crop species as the arithmetic sum (count) of tree crop species, where all unidentified tree crop species are just counted as one tree crop species. We proceed similarly for planted forest species. The difference is made between tree crops and planted forests because plantations of tree crops are nearer the populations and more accessible to them for livelihood subsistence

than forest plantations. These two simple metrics of diversification of tree species indicate that most sub-national regions within SSA have zero or one species of tree crops and planted forest in the tree plantations. In Eastern and Southern Africa (ESA), sub-national regions with diversity scores of tree crops and planted forests of just one species have similar means of gendered GNIs to those regions with zero diversity scores (Figure 3); however, the mean of HDI by gender in regions with zero species is relatively lower compared to regions with one tree species (Figure 4). In Western and Central Africa (WCA), while the diversification of forest plantations seems to make no difference in terms of mean values of GNI and HDI by gender, subnational regions with a tree crop diversity score of value one have on average higher HDI and GNI mean values by gender than sub-national regions with zero tree crop diversity score (Figure 4 and Figure 5). In ESA or WCA, Figure 6 shows no differences in IWI between zero and one tree species in the tree plantations; in WCA, subnational regions with mono-species tree plantations have larger IWI than multispecies tree plantations.



GNI: Gross National Income per capita in thousands of USD, (2011 PPP).
ESA: Eastern and Southern Africa. WCA: Western and Central Africa.
Distribution of diversity score of tree crop species:
ESA --- zero (95%), one (5%); WCA --- zero (52.0%), one (43%), two or more (5%).
Distribution of diversity of planted forest species:
ESA --- zero (75%), one (25%); WCA --- zero (63%), one (24%), two or more (13%).

Figure 4: Mean and 95% confidence interval of subnational-year GNI by gender, diversity score of tree species, and SSA regions

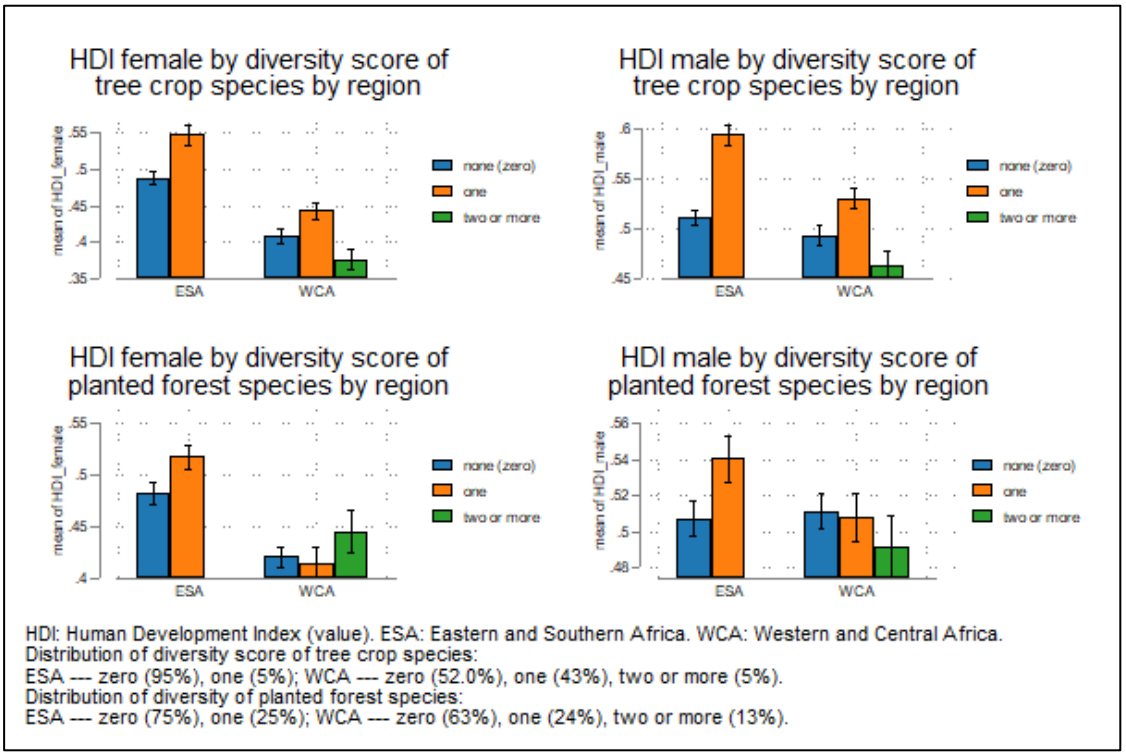


Figure 5: Mean and 95% confidence interval of subnational-year HDI by gender, diversity score of tree species, and SSA regions

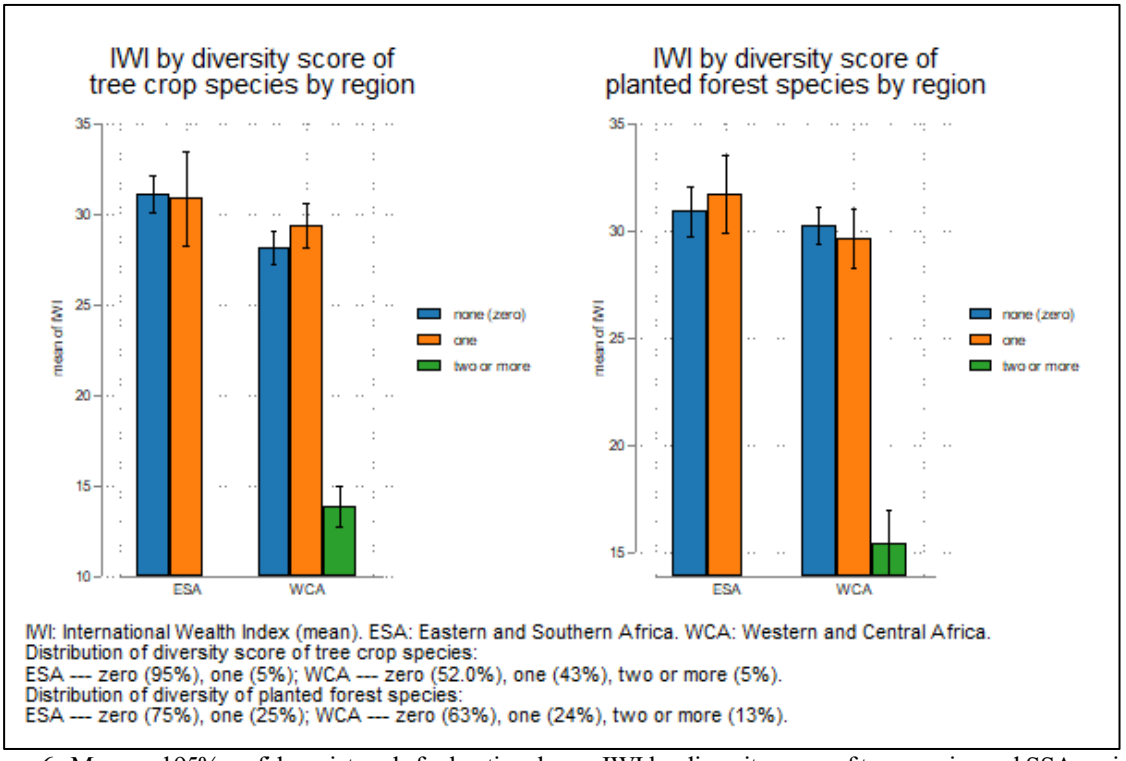


Figure 6: Mean and 95% confidence interval of subnational-year IWI by diversity score of tree species and SSA regions

1.2 Methods

3.1.1 *Computation of the binary tree planting treatment variable*

Our analysis draws on a panel of N_i countries, each with N_{ij} sub-national divisions or regions observed

for T years. We denote $Y_{ij,t}$ the human well-being outcome, which is observed for each region $j \in \{1, \dots, N_{ij}\}$ of country $i \in \{1, \dots, N_i\}$ in year $t \in \{1, \dots, T\}$. We observe the area under new tree plantations in each country-region-year in square kilometers $A_{ij,t}$, as sourced from the global map of planting years of plantations dataset.[5] $A_{ij,t}$ can take the value zero in any year t where the region ij has not increased their tree plantation area; however, this does not mean that the region ij has no tree plantations in year t because $A_{ij,t-k}$ could be nonzero for $1 \leq k \leq t-1$, in which case the actual area under tree plantations in year t is the nearest precedent nonzero $A_{ij,t-k}$. Moreover, when $A_{ij,t}$ takes a nonzero value a, it could be the case that $A_{ij,t-1} = a$ if the region has simply expanded the density of trees in the same area, in which case the actual area under tree plantations in year t is $A_{ij,t}$. In cases when $A_{ij,t}$ takes a nonzero value a and $A_{ij,t-1}$ takes a nonzero value $b \neq a$, we have either an extension of the actual area under tree plantations ($b < a$) or a clearing ($b > a$). Based on this understanding, we denote $\tilde{A}_{ij,t}$ the actual area under tree plantations in year t and compute it as the nearest precedent nonzero $A_{ij,t-k}$ if $A_{ij,t} = 0$; else, $\tilde{A}_{ij,t} = A_{ij,t}$.

Given that tree planting is a long-term investment, its effect on human well-being in year t accrues not just to $\tilde{A}_{ij,t}$ but from ℓ precedent years, which we set to three or four years, and denote D_ℓ our binary treatment variable computed as follows. The triennial treatment D_3 takes value one if the actual area under tree plantations has increased within the last three years, *i.e.* if $\tilde{A}_{ij,t} - \tilde{A}_{ij,t-1} > 0$ or $\tilde{A}_{ij,t-1} - \tilde{A}_{ij,t-2} > 0$ or $\tilde{A}_{ij,t-2} - \tilde{A}_{ij,t-3} > 0$, and value zero otherwise. The quadrennial treatment D_4 takes value one if $D_3 = 1$ or if $\tilde{A}_{ij,t-3} - \tilde{A}_{ij,t-4} > 0$, and value zero otherwise. Our main hypothesis is that, on average, in a regional population, there are human well-being benefits from improvements in the actual area under tree plantations within ℓ years:

$$\mathrm{E}\left[Y_{ijt} | D_\ell = 1\right] - \mathrm{E}\left[Y_{ijt} | D_\ell = 0\right] > 0\,, \qquad \text{(Eq.1)}$$

where $\mathrm{E}[\cdot]$ is the expectation operator.

3.1.2 *Estimation of the effects of tree planting on human well-being*

We test our main hypothesis by estimating the following Multi–Way Fixed Effect (Correia, 2016) OLS regression model:

$$Y_{ij,t} = \alpha_i + \beta_j + \gamma_t + \boldsymbol{X}'_{ij,t}\lambda + \tau * D_\ell + \varepsilon_{ij,t}, \qquad \text{(Eq.2)}$$

where α_i are country-fixed effects, β_j are region fixed effects, γ_t are year fixed effects, and $X_{ij,t}$ is a vector of covariates that may influence both D_ℓ and $Y_{ij,t}$. Because institutional characteristics such as female representation in the parliament and socioeconomic characteristics such as age at reproduction and education may influence human development, income, employment, health, and/or tree planting, the vector $X_{ij,t}$ includes the average age at first birth of women aged between 20 and 50 years old in the region ij in year t, the average number of schooling years in the region ij in year t, and the share of seats in parliament held by women in the country i in year t. Regional population weights are used in estimating Eq.2 because the regions are not alike across countries; certain regions of countries are as populous as other countries. Controlling for these covariates, the time-invariant country and region effects, and the year-fixed effects, differences in human well-being outcomes $Y_{ij,t}$ are due to differences in the treatment status D_ℓ. Following Eq.1, we expect $\tau > 0$, meaning the increase in tree planting area at the extensive margin within an ℓ–year time window raises $Y_{ij,t}$ by τ units.

However, it could be argued that increases in the area under tree plantations are endogenous in Eq.2 due to three possible concerns. First, there are measurement errors in the spatial metrics of the area under tree plantation (Du et al., 2022; Jain, 2020), which can bias the estimated treatment effects. Second, if tree plantation areas are expanded using oil palm trees --- the main tree species in the data --- due to the suitability of the subnational regions to oil palm trees, then omitting variables such as the choice of tree

[5] A global map of planting years of plantations (Du et al., 2022).

species planted would bias the estimated treatment effects. Third, it could be the case that subnational regions with better human well-being outcomes are those enlarging tree plantation areas, thus bringing reverse causality in the relationship between the treatment and the outcomes. Nevertheless, we disregard the presumption that wealthier countries or regions systematically plant more trees compared to poorer countries or regions for several reasons. Many countries across different income levels have implemented tree planting programs considering the decreasing biodiversity, carbon storage, and economic incomes caused by forest loss (Du et al., 2022). Empirical evidence also suggests notable tree planting initiatives by rural households in developing countries, as a proactive approach towards leveraging tree resources for economic and environmental benefits, irrespective of the countries' income levels (Ding et al., 2017). Land availability is often more of a constraint for tree planting interventions than financial resources. Land area available for tree planting in some countries is much lower than reforestation commitments, making it more challenging to expand tree plantation areas (Brancalion and Holl, 2020).

To account for the remaining sources of endogeneity, we instrument increases in the area under tree plantations using (1) the previous share of neighboring subnational regions with increases in the area under tree plantations and (2) the interaction of the previous share of neighbor subnational regions with increases in the area under tree plantations and the mean oil palm suitability index for the subnational region of interest. In the computation of the first instrument, a neighboring subnational region to a subnational region of interest is defined as being a subnational region of a country bordering the country of the subnational region of interest or as being another subnational region of the country of the subnational region of interest. When the country of the subnational region of interest has no bordering country in the study sample (cases of Kenya, Lesotho, and Namibia), we use as neighbors all subnational regions of the sample other than the subnational region of interest. Thus, the previous share of neighbor subnational regions with increases in the area under tree plantations within four years for the subnational region "Atacora + Donga" of Benin in the year 1992 is the average value of D_4 in the year 1991 across the 5 subnational regions of Benin other than "Atacora + Donga" and all 6 subnational regions of Togo. The previous share of neighbor subnational regions with increases in the area under tree plantations within four years for the subnational region "Central" of Kenya in the year 2016 is the average value of D_4 in the year 2015 across 102 subnational regions of the sample, including the seven subnational regions of Kenya other than "Central" and all 95 subnational regions of the 11 countries of the sample other than Kenya. This first instrument addresses the concerns about measurement errors in the treatment status of a subnational region of interest because the computation of the instrument excludes the subnational region of interest and is based on the lagged values of the treatment variable in nearest subnational regions. Positive values of this first instrument are expected to be associated with a higher probability of increasing tree plantation areas.

The second instrument addresses the concerns about the choice of tree species planted. By interacting the first treatment with the mean oil palm suitability index for the subnational region of interest, the second instrument becomes more relevant for the tree planting treatment in subnational regions where oil palm trees may be the tree species planted. In that regard, the first instrument becomes more relevant for the tree planting treatment in subnational regions where oil palm trees may not be the tree species planted. Because the values of the suitability index are in decreasing order (Strona et al, 2018 and Figure A2), lower values of the second instrument are expected to be associated with a higher probability of increasing tree plantation areas, especially in subnational regions more suitable to oil palm trees.

The validity of spatial instruments has been reviewed in the literature, with two main criticisms raised, namely (i) simultaneity between the instrument and the endogenous treatment variable; and (ii) spillover and interdependent outcomes possibly violating the exclusion restriction of the spatial instruments (Betz et al, 2018). We avoid the simultaneity criticism by using lagged spatial instruments rather than the contemporaneous share of neighboring subnational regions with increases in the area under tree plantations. Following Reed (2015), if the lagged spatial instrument is correlated with the endogenous treatment as evidenced in the first-stage regressions for the instrument variables method (Appendix Tables A2-A4), we avoid simultaneity bias. To address the second criticism, since the human well-being outcome of a subnational region of interest is likely to be correlated with the mean outcome of their neighboring subnational regions, we need to ensure that the latter is not correlated with the spatial

instrument. We present the evidence on the interdependence between the spatial instrument, the mean outcomes of neighbors, and the outcome in Table A5 for the triennial and quadrennial treatments. While subnational regions do have interdependent outcomes for GNI (Panel B, columns one and four) and IWI (Panel B, columns three and six), Panel A shows no apparent interdependence between the spatial instrument and the mean outcomes for GNI (columns one and four) and IWI (columns three and six). Panels A and B also indicate no apparent interdependence between the spatial instrument, the mean HDI of neighbors, and the HDI of a subnational region (columns two and five). Thus, our spatial instrument is not likely to influence the human well-being outcome of a subnational region through the neighboring subnational regions.

3.1.3 *Mediation analysis*

To empirically examine the hypotheses shown in Figure 1, we use a two-step approach to apprehend the pathways to the impact of tree planting where it exists following Acharya *et al.* (2016). First, we augment the regression Eq.2 above with potential mediating factor(s), which fall under employment, environment, and tree species diversification secondary outcomes as shown in Figure 1. If the individual effect of a secondary outcome on human well-being is significant, then that secondary outcome is a mediator of the impact of tree planting on human well-being. If the interaction of the treatment and a secondary outcome has a significant effect on human well-being in the augmented regression, then that secondary outcome is a moderator of the impact of tree planting on human well-being. Else, if a secondary outcome has insignificant individual and interaction effects on human well-being in the augmented regression, then that secondary outcome is irrelevant to the impact pathways. At this step, there may be multiple mediators and moderators of the impact of tree planting on human well-being.

Second, we subtract the effect of a subset of these mediators and moderators from the dependent variable to obtain what is called the "de-mediated" dependent variable. By regressing the latter on the treatment and the independent variables used in Eq.2, we are able to conclude if the subset whose effect is subtracted is the pathway at play. To validate the subset of mediators and moderators, the effect of tree planting on the "de-mediated" human well-being should be insignificant. In other terms, after removing the effects of this subset of mediators and moderators, there is no additional effect of tree planting, which means the total effect of tree planting goes through the validated mediator(s) and moderator(s).

3.1.4 *Estimation of the effects of tree species diversification on human well-being*

Because the local communities may be constrained by the area allocated to trees and simply choose to raise or lower the density of trees in the same area over time, the diversity of tree species instead of the area under tree planting may also reveal the impact of tree planting on the targeted human well-being outcomes, including GNI, HDI, and IWI. We examine the potential relationship between tree species diversification in tree plantations and human well-being outcomes by estimating Eq.3 using a Multi–Way Fixed Effect OLS regression model:

$$Y_{ij,t} = \alpha_i + \beta_j + \gamma_t + X'_{ij,t}\lambda + \sum_{s=1}^{\overline{s}} \delta_s * I\left[S_{ij,t} = s\right] + \sum_{p=1}^{\overline{p}} \theta_p * I\left[P_{ij,t} = p\right] + \epsilon_{ij,t}, \quad \text{(Eq.3)}$$

where $S_{ij,t}$ stands for the diversity score (count) of tree species in tree crop plantations ranging from $s = 0$ to $s = \overline{s}$, and $P_{ij,t}$ stands for the diversity score (count) of tree species in forest plantations ranging from $p = 0$ to $p = \overline{p}$, and $I[\cdot]$ is an indicator variable, and the parameters δ_s and θ_p are the coefficients of interest. Since $s = 0$ and $p = 0$ are omitted in Eq.3, δ_s captures the effect of value s of $S_{ij,t}$ on $Y_{ij,t}$ relative to zero tree crop species; similarly, θ_p captures the effect of value p of $P_{ij,t}$ on $Y_{ij,t}$ relative to zero planted forest species. We expect $\delta_s > 0$ and $\theta_p > 0$ to indicate well-being benefits from tree species diversification.

4 Results

4.1 Does tree planting improve human well-being at the sub-national level? And where?

We present results on the three targeted human well-being indicators, GNI, HDI, and IWI, using the triennial and quadrennial treatment variables. Additional robustness checks done using a quinquennial

treatment variable are presented in Supplemental file (Section 1.1 – Tables S1-S6).

4.1.1 *Effects of tree planting on human well-being outcomes at the sub-national level across sub-Saharan Africa*

Table 2 presents the main results for the full sample using OLS (Panel A) and IV (Panel B). IV estimation results indicate the endogeneity of the tree planting treatment at the 1% critical level in all columns except those for HDI (columns 2 and 4). The failure of rejection of the null hypothesis that the treatment is not endogenous in HDI regressions suggests the reliance upon the OLS estimates for HDI (Panel A, columns 2 and 4) but the IV estimates for GNI (Panel B, columns 1 and 4) and IWI (Panel B, columns 3 and 6). In addition, the IV estimations are not overidentified based on the Chi-square test, and the first-stage regression results in Table A.2 show significant instruments in all regressions, and the relevance of the instruments for the treatment is relatively strong for both the quadrennial treatment in Panel B (Kleibergen-Paap F-stat = 15.239) and the triennial treatment in Panel A (Kleibergen-Paap F-stat 11.069). Thus, interpreting IV estimates in GNI and IWI columns is appropriate.

The IV estimates suggest that tree planting has positive and significant effects on subnational GNI per capita and IWI. Specifically, we find that increases in area under tree plantation over a three-year period lead to an increase in subnational GNI per capita by approximately 54% (Panel B, column 1), while increases in area under tree plantation over a four-year period raise subnational GNI per capita by approximately 45% (Panel B, column 4).[6] Furthermore, our findings reveal that an increase in the area under tree plantation at the extensive margin within a three-year time window is associated with a 12-unit improvement of the subnational wealth index, which is an increase in subnational IWI by 44% (Panel B, column 3), while an increase in the area under tree plantation at the extensive margin within a four-year time window is associated with a 10-unit improvement of the subnational wealth index, which is an increase in subnational IWI by 37% (Panel B, column 6).[7]

Notwithstanding, the results reveal the absence of improvement in subnational HDI, either with the triennial or quadrennial treatment (Panel A, columns 2 and 5).

[6] The effect size is (1.18510 / 2.192) * 100 = 54.06% for the triennial treatment and (0.97428 / 2.172) * 100 = 44.86% for the quadrennial treatment.

[7] The effect size is (11.91457 / 27.21) * 100 = 43.79% for the triennial treatment and (9.99384 / 27.20) * 100 = 36.74% for the quadrennial treatment.

Table 2: Multi–way fixed effects OLS and IV estimation results for the impact of tree planting on human well-being outcomes in sub-Saharan Africa

Panel A: OLS estimations	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Increased area under tree planting within the last 3 years (triennial)	0.14362***	-0.00165	1.41305***			
	(0.04773)	(0.00263)	(0.52244)			
Increased area under tree planting within the last 4 years (quadrennial)				0.16072***	-0.00268	1.55829***
				(0.05643)	(0.00307)	(0.50602)
Mean age at first birth of women aged 20-50	0.19445**	-0.00663	1.88313*	0.19696**	-0.00658	1.90986*
	(0.07681)	(0.00430)	(1.02892)	(0.07735)	(0.00430)	(1.02363)
Mean years schooling (years)	0.09485*	0.01530***	-0.67038	0.09501*	0.01527***	-0.66947
	(0.05331)	(0.00293)	(0.69735)	(0.05370)	(0.00293)	(0.70650)
Share of seats in parliament, female (percent held by women)	0.00575	0.00287***	0.07066	0.00606	0.00286***	0.07360
	(0.00525)	(0.00043)	(0.05192)	(0.00526)	(0.00042)	(0.05165)
Observations	2,253	2,253	2,253	2,253	2,253	2,253
R-squared	0.97314	0.95783	0.95049	0.97313	0.95785	0.95044
Tree planting treatment	Triennial	Triennial	Triennial	Quadrennial	Quadrennial	Quadrennial
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable in the absence of treatment	2.192	0.431	27.21	2.172	0.427	27.20
Testing tree planting effect = 0, p-value =	0.00327	0.532	0.00796	0.00527	0.383	0.00263
Panel B: IV estimations	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Increased area under tree planting within the last 3 years (triennial)	1.18510***	0.04897*	11.91457***			
	(0.37469)	(0.02705)	(4.14091)			
Increased area under tree planting within the last 4 years (quadrennial)				0.97428***	0.03240	9.99384***
				(0.34229)	(0.02398)	(3.61966)
Mean age at first birth of women aged 20-50	0.07409	-0.01304**	0.75142	0.12391	-0.01026**	1.23397
	(0.09084)	(0.00566)	(1.15757)	(0.08396)	(0.00495)	(1.06555)
Mean years schooling (years)	0.12485**	0.01674***	-0.35569	0.11660**	0.01619***	-0.43334
	(0.05520)	(0.00353)	(0.65946)	(0.05419)	(0.00327)	(0.68010)
Share of seats in parliament, female (percent held by women)	0.01058**	0.00310***	0.11765**	0.01101**	0.00307***	0.12311**
	(0.00470)	(0.00053)	(0.05134)	(0.00471)	(0.00051)	(0.05020)
Observations	2,253	2,253	2,253	2,253	2,253	2,253
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap F-stat	11.069	11.069	11.069	15.239	15.239	15.239
Overidentification test of all instruments: Chi-square=	0.915	3.682	0.543	0.239	5.134	0.155
Overidentification test of all instruments: p-value=	0.339	0.055	0.461	0.625	0.023	0.694
Endogeneity test of endogenous regressors: Chi-square=	11.146	1.506	10.941	7.578	0.395	7.444
Endogeneity test of endogenous regressors: p-value=	0.001	0.220	0.001	0.006	0.530	0.006

Notes: This table reports the estimation results for the effects of tree planting on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP), and the Human Development Index (HDI) in sub-Saharan Africa using country-region-year fixed effects. Country-region-year observations cover twelve sub-Saharan African countries over the period of 1992 through 2016. The estimation sample includes five Eastern and Southern African countries (Kenya, Lesotho, Mozambique, Namibia, and Zimbabwe) and seven Western and Central African countries (Benin, Cote d'Ivoire, Ghana, Guinea, Togo, Congo Democratic Republic, and Rwanda). Significance levels: *** 1%, ** 5%, * 10%.

4.1.2 *Dis-aggregated effects within sub-Saharan Africa*

To gain insight into our main findings through a heterogeneity analysis, we estimate differential treatment effects between Western and Central Africa (WCA) and Eastern and Southern Africa (ESA). These two subregions of Africa are diverse in geography, culture, and economy. OLS and IV effects of increasing the area under tree planting on the main outcomes are presented for WCA countries in Table 3 and for ESA countries in Table 4. In the regressions for WCA, the endogeneity test results suggest that IV estimates are more consistent than the OLS estimates in the GNI and IWI columns of Table 3, with the instruments being relatively strong for both the triennial and quadrennial treatments (Kleibergen-Paap F-stat > 19 in Panels A and B of Table A.3). In the regressions for ESA, the relevance of the instruments also remains satisfactory (Kleibergen-Paap F-stat > 15 in Panels A and B of Table A.4), but we fail to reject the endogeneity of the treatment variables only in the IWI regressions (Table 4, columns 3 and 6). Consequently, we rely on IV estimates for per-capita GNI and IWI in Table 3 and for IWI in Table 4.

Findings from Table 3 in WCA indicate that an extensive-margin increase in the area of tree plantations over the last three or four years has positive effects on subnational per-capita GNI (Panel B, columns 1 and 4), with respective sizes of 41% and 31% for the triennial and quadrennial treatments.[8] The rise in subnational per capita incomes due to the tree planting treatment in WCA is 184 to 319 USD per year. We also observed a rise in subnational IWI by 45% over three years and 39% over four years (Panel B, columns 3 and 6)[9], but no effects on subnational HDI (Panel A, columns 2 and 5).

Focusing on ESA (Table 4), we find insignificant tree planting effects on subnational HDI (Panel A, columns 2 and 5). In addition, there are insignificant tree planting effects on subnational GNI (Panel A, columns 1 and 4). We find, however, evidence of positive effects on subnational IWI, which are significant at the 1% and 5% critical levels, respectively for the triennial and quadrennial treatments (Panel B, columns 3 and 6) and are about half of the magnitudes observed for the effects on subnational IWI in WCA.

[8] The effect size is (0.95803 / 2.334) * 100 = 41.05% for the triennial treatment and (0.73608 / 2.350) * 100 = 31.32% for the quadrennial treatment.

[9] The effect size is (13.02017 / 28.92) * 100 = 45.02% for the triennial treatment and (11.42532 / 29.15) * 100 = 39.19% for the quadrennial treatment.

Table 3: Multi–way fixed effects OLS and IV estimation results for the impact of tree planting on human well-being outcomes in Western and Central Africa

Panel A: OLS estimations	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Increased area under tree planting within the last 3 years (triennial)	0.20857***	-0.00075	1.98212***			
	(0.05047)	(0.00255)	(0.65414)			
Increased area under tree planting within the last 4 years (quadrennial)				0.23567***	-0.00322	2.12077***
				(0.05598)	(0.00267)	(0.55091)
Mean age at first birth of women aged 20-50	0.21442**	-0.00750*	2.46931**	0.21623**	-0.00736*	2.49420**
	(0.08403)	(0.00433)	(1.14346)	(0.08510)	(0.00431)	(1.13999)
Mean years schooling (years)	0.00842	0.01619***	-1.53809*	0.00540	0.01618***	-1.56789*
	(0.06412)	(0.00415)	(0.84446)	(0.06527)	(0.00413)	(0.86235)
Share of seats in parliament, female (percent held by women)	0.00221	0.00355***	0.05598	0.00290	0.00353***	0.06155
	(0.00537)	(0.00028)	(0.05005)	(0.00533)	(0.00028)	(0.04947)
Observations	1,137	1,137	1,137	1,137	1,137	1,137
R-squared	0.96935	0.96657	0.94557	0.96926	0.96666	0.94531
Tree planting treatment	Triennial	Triennial	Triennial	Quadrennial	Quadrennial	Quadrennial
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable in the absence of treatment	2.334	0.439	28.92	2.350	0.436	29.15
Testing tree planting effect = 0, p-value =	0.000123	0.771	0.00372	9.54e-05	0.232	0.000311
Panel B: IV estimations	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Increased area under tree planting within the last 3 years (triennial)	0.95803***	-0.03918***	13.02017***			
	(0.31461)	(0.01443)	(3.76440)			
Increased area under tree planting within the last 4 years (quadrennial)				0.73608**	-0.05614***	11.42532***
				(0.28923)	(0.01833)	(3.23797)
Mean age at first birth of women aged 20-50	0.15577*	-0.00411	1.32866	0.18899**	-0.00370	1.68329
	(0.09053)	(0.00418)	(1.29797)	(0.08724)	(0.00403)	(1.21506)
Mean years schooling (years)	0.02606	0.01521***	-1.25601	0.00903	0.01566***	-1.47389*
	(0.05904)	(0.00426)	(0.79511)	(0.06171)	(0.00423)	(0.83460)
Share of seats in parliament, female (percent held by women)	0.00651	0.00333***	0.11405**	0.00693	0.00310***	0.13126**
	(0.00537)	(0.00028)	(0.05537)	(0.00514)	(0.00028)	(0.05269)
Observations	1,137	1,137	1,137	1,137	1,137	1,137
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap F-stat	19.816	19.816	19.816	24.402	24.402	24.402
Overidentification test of all instruments: Chi-square=	0.920	8.531	0.171	1.463	6.084	0.140
Overidentification test of all instruments: p-value=	0.338	0.003	0.680	0.226	0.014	0.708
Endogeneity test of endogenous regressors: Chi-square=	8.298	0.373	14.468	5.846	1.280	11.157
Endogeneity test of endogenous regressors: p-value=	0.004	0.542	0.000	0.016	0.258	0.001

Notes: This table reports the estimation results for the effects of tree planting on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP), and the Human Development Index (HDI) in Western and Central Africa using country-region-year fixed effects. Country-region-year observations cover seven Western and Central African countries (Benin, Cote d'Ivoire, Ghana, Guinea, Togo, Congo Democratic Republic, and Rwanda) over the period of 1992 through 2016. Significance levels: *** 1%, ** 5%, * 10%.

Table 4: Multi–way fixed effects OLS and IV estimation results for the impact of tree planting on human well-being outcomes in Eastern and Southern Africa

Panel A: OLS estimations	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Increased area under tree planting within the last 3 years (triennial)	0.07347	-0.00065	1.15167**			
	(0.08377)	(0.00726)	(0.57076)			
Increased area under tree planting within the last 4 years (quadrennial)				0.06905	-0.00081	1.33580*
				(0.09800)	(0.00787)	(0.71056)
Mean age at first birth of women aged 20-50	0.12767	-0.01338	0.65111	0.13174	-0.01339	0.67425
	(0.13615)	(0.00868)	(1.51454)	(0.13533)	(0.00861)	(1.50113)
Mean years schooling (years)	0.50600***	0.02134***	3.67773***	0.50505***	0.02134***	3.68269***
	(0.09300)	(0.00650)	(0.79252)	(0.09291)	(0.00652)	(0.79876)
Share of seats in parliament, female (percent held by women)	0.00903**	0.00144***	-0.03633	0.00920**	0.00144***	-0.03257
	(0.00414)	(0.00024)	(0.05182)	(0.00411)	(0.00024)	(0.05241)
Observations	1,116	1,116	1,116	1,116	1,116	1,116
R-squared	0.98049	0.96392	0.97326	0.98048	0.96392	0.97331
Tree planting treatment	Triennial	Triennial	Triennial	Quadrennial	Quadrennial	Quadrennial
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable in the absence of treatment	2.074	0.424	25.77	2.038	0.420	25.73
Testing tree planting effect = 0, p-value =	0.385	0.929	0.0489	0.484	0.919	0.0658
Panel B: IV estimations	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Increased area under tree planting within the last 3 years (triennial)	0.18152	-0.00021	5.99820***			
	(0.20757)	(0.01986)	(2.12434)			
Increased area under tree planting within the last 4 years (quadrennial)				0.17458	0.00352	6.10414**
				(0.23706)	(0.02294)	(2.57531)
Mean age at first birth of women aged 20-50	0.09747	-0.01556	-0.18774	0.10687	-0.01620	0.06617
	(0.14029)	(0.01029)	(1.46407)	(0.13998)	(0.01011)	(1.45762)
Mean years schooling (years)	0.51349***	0.02145***	4.20706***	0.51140***	0.02174***	4.16431***
	(0.10182)	(0.00704)	(0.94654)	(0.10065)	(0.00714)	(0.94275)
Share of seats in parliament, female (percent held by women)	0.00918**	0.00141***	-0.03653	0.00964**	0.00142***	-0.02011
	(0.00426)	(0.00024)	(0.06050)	(0.00426)	(0.00025)	(0.06125)
Observations	1,116	1,116	1,116	1,116	1,116	1,116
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap F-stat	21.967	21.967	21.967	15.573	15.573	15.573
Overidentification test of all instruments: Chi-square=	3.553	2.281	2.185	3.354	2.496	2.475
Overidentification test of all instruments: p-value=	0.059	0.131	0.139	0.067	0.114	0.116
Endogeneity test of endogenous regressors: Chi-square=	0.952	0.069	8.455	0.723	0.293	7.667
Endogeneity test of endogenous regressors: p-value=	0.329	0.793	0.004	0.395	0.589	0.006

Notes: This table reports the estimation results for the effects of tree planting on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP), and the Human Development Index (HDI) in Eastern and Southern Africa using country-region-year fixed effects. Country-region-year observations cover five Eastern and Southern African countries (Kenya, Lesotho, Mozambique, Namibia, and Zimbabwe) over the period of 1992 through 2016. Significance levels: *** 1%, ** 5%, * 10%.

4.2 Does the diversification of tree species matter for human well-being?

One thing is to plant trees for better human well-being, and another thing is to plant the right trees for better human well-being because people could be sourcing their livelihoods one way or the other from certain species in the tree plantations, but not others. To explore the association between the diversification of tree species and human well-being outcomes, we describe the distribution of gendered human well-being outcomes (GNI and HDI) by species of tree crops and planted forests in the years 2000 through 2016 and estimate the effects of diversity scores of tree crops and planted forests on the gendered human well-being outcomes.

In the pooled sub-national sample for those years (2000-2016), we find that the medians of GNI female and male are higher in regions planting unidentified species (tree crops or planted forests) than those in other regions planting no species or the identified species (Figure 7). The same applies to HDI male and HDI female in regions planting unidentified species (Figure 8). Thus, certain tree species, when planted as crops or forests, have the potential to uplift income and human development for women as well as men. Because the proportion of regions with unidentified species is low (2% for tree crops and 25% for planted forests) relative to the proportion of regions without tree crop plantations (73%) and forest plantations (69%), means of human well-being outcomes are skewed towards the regions without tree plantations.

Identifying locally these unknown species that associate with higher GNI and HDI and intensifying them in the regions where they are marginal or where they are absent could improve human well-being. This is largely the case in Eastern and Southern Africa, where the regions with unidentified species (tree crops or planted forests) have less variable and higher median gendered GNI and HDI than the other regions (Figure A.3 and Figure A.4). In Western and Central Africa, regions with monospecies plantations of oil palm trees are associated with higher medians of gendered GNI and HDI than the other regions (Figure A.5 and Figure A.6); regarding planted forest species, it is unclear whether the regions with better human well-being outcomes are those with unidentified species, with at least two identified species, or without forest plantations (Figure A.5 and Figure A.6).

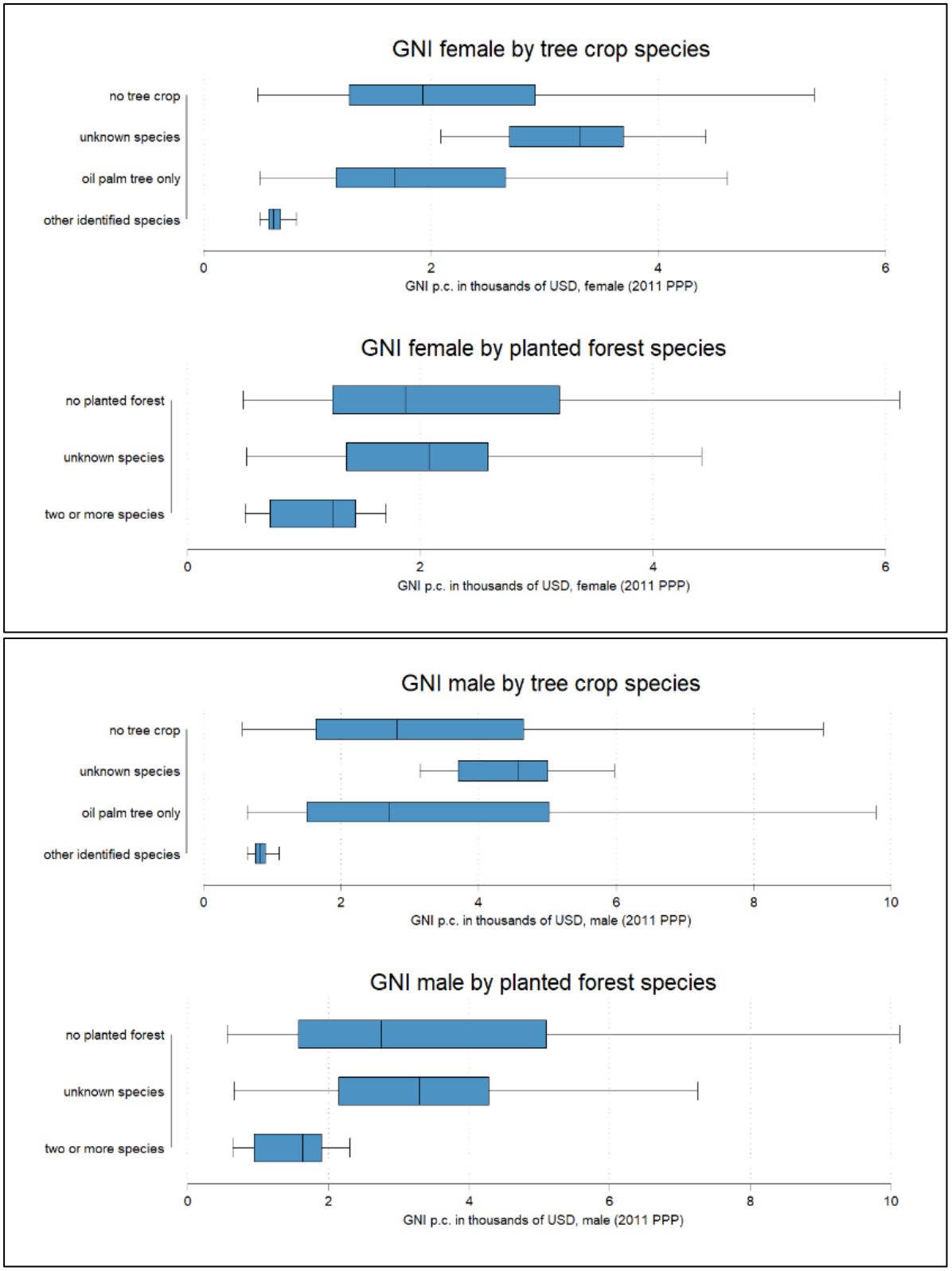


Figure 7: Distribution of subnational GNI female and male by tree species in the pooled sample

Notes: Outside values of box plots are excluded. The distribution of diversity of tree crop species in the country-region-year pooled sample is: no tree crop (73%), unknown species (2%), oil palm tree only (22%), other identified species (with or without oil palm trees) (3%). The last category includes: rubber trees only; palm trees only; oil palm and rubber trees; oil palm and palm trees; oil palm and coconut trees; oil palm, coconut, and rubber trees. The distribution of diversity of planted forest species in the country-region-year pooled sample is: no planted forest (69%), unknown species (25%); and two or more species (6%). The last category includes: East Asian white birch and unknown species; Bocote and unknown species; East Asian white birch, pine, and unknown species; East Asian white birch, eucalyptus, and unknown species; East Asian white birch, pine, cypress, and unknown species; East Asian white birch, eucalyptus, cypress, and unknown species; East Asian white birch, eucalyptus, pine, and unknown species; East Asian white birch, eucalyptus, cypress, pine, and unknown species.

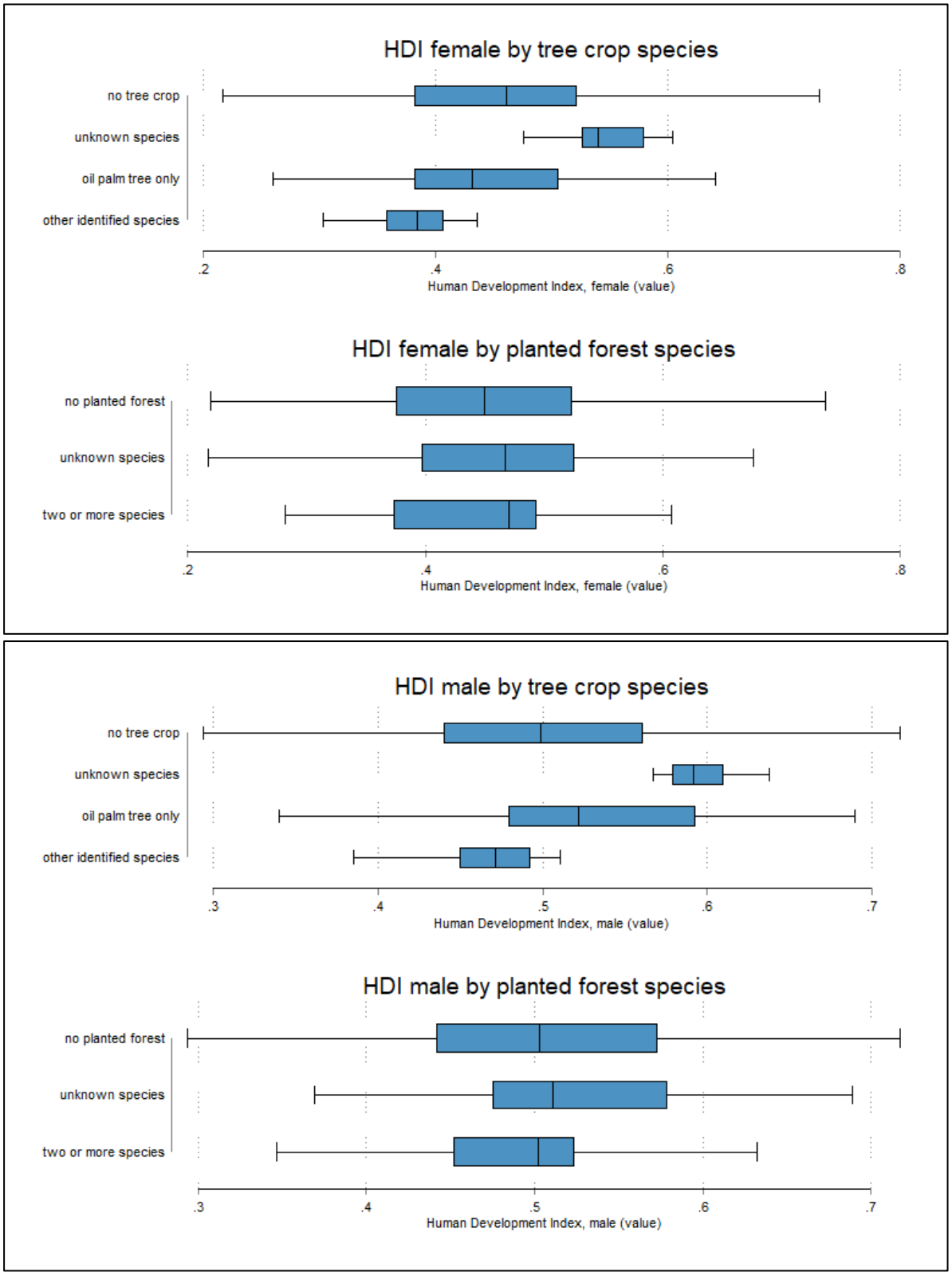


Figure 8: Distribution of subnational HDI female and male by tree species in the pooled sample

Notes: Outside values of box plots are excluded. The distribution of diversity of tree crop species in the country-region-year pooled sample is: no tree crop (73%), unknown species (2%), oil palm tree only (22%), other identified species (with or without oil palm trees) (3%). The last category includes: rubber trees only; palm trees only; oil palm and rubber trees; oil palm and palm trees; oil palm and coconut trees; oil palm, coconut, and rubber trees. The distribution of diversity of planted forest species in the country-region-year pooled sample is: no planted forest (69%), unknown species (25%); and two or more species (6%). The last category includes: East Asian white birch and unknown species; Bocote and unknown species; East Asian white birch, pine, and unknown species; East Asian white birch, eucalyptus, and unknown species; East Asian white birch, pine, cypress, and unknown species; East Asian white birch, eucalyptus, cypress, and unknown species; East Asian white birch, eucalyptus, pine, and unknown species; East Asian white birch, eucalyptus, cypress, pine, and unknown species.

Estimation results for the effects of tree species diversity scores on the GNI per capita and HDI by gender as well as IWI are presented across the twelve SSA countries in Table 5, and separately for Eastern and Southern Africa countries (Panel A) and for Western and Central Africa countries (Panel B) in Table 6.

Table 5: Multi–way fixed effects OLS estimation results for the impact of diversity scores of tree species on income and human development at the subnational-year level in sub-Saharan Africa

VARIABLES	(1) GNI female	(2) GNI male	(3) GNI	(4) HDI female	(5) HDI male	(6) HDI	(7) IWI
Diversity score (count) of tree crop species: one	0.10285	0.29808**	0.25536**	-0.00185	-0.00223	0.00066	2.35482**
	(0.09200)	(0.14716)	(0.10661)	(0.00381)	(0.00264)	(0.00348)	(0.95693)
Diversity score (count) of tree crop species: two or more	0.08441	0.34940**	0.23142*	-0.00362	-0.00378	0.00009	2.14900**
	(0.09706)	(0.15583)	(0.12601)	(0.00422)	(0.00304)	(0.00514)	(1.01673)
Diversity score (count) of planted forest species: one	0.08142	0.14000	0.17613**	0.00037	0.00065	-0.00073	1.72753**
	(0.06319)	(0.09528)	(0.07798)	(0.00302)	(0.00223)	(0.00325)	(0.76616)
Diversity score (count) of planted forest species: two or more	0.10330**	0.19109*	0.21748***	-0.00213	0.00023	0.01058	0.31539
	(0.04789)	(0.10026)	(0.08109)	(0.00473)	(0.00408)	(0.01328)	(0.87388)
Mean age at first birth of women aged 20-50	0.17590**	0.37213***	0.20799***	0.00090	-0.00491	-0.00746*	2.05601**
	(0.08229)	(0.13336)	(0.07820)	(0.00603)	(0.00534)	(0.00432)	(1.01741)
Mean years schooling (years)	0.03349	0.09240	0.09857*	0.01598***	0.01946***	0.01557***	-0.66905
	(0.04235)	(0.06948)	(0.05353)	(0.00302)	(0.00263)	(0.00295)	(0.71530)
Share of seats in parliament, female (percent held by women)	0.01383***	-0.00373	0.00633	0.00092***	0.00102***	0.00287***	0.07246
	(0.00424)	(0.00760)	(0.00520)	(0.00030)	(0.00027)	(0.00043)	(0.05111)
Observations	943	943	2,253	943	943	2,253	2,253
R-squared	0.97893	0.98237	0.97320	0.98197	0.98229	0.95802	0.94800
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable with zero tree crop species	2.045	2.912	2.362	0.447	0.509	0.450	28.05
Mean dependent variable with zero planted forest species	2.068	3.268	2.413	0.438	0.512	0.446	30.22

Notes: This table reports the estimation results for the effects of diversity scores of tree species on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP) and the Human Development Index (HDI) by gender in sub-Saharan Africa using country-region-year fixed effects. Country-region-year observations cover twelve sub-Saharan African countries over the period of 1992 through 2016. The estimation sample includes five Eastern and Southern African countries (Kenya, Lesotho, Mozambique, Namibia, and Zimbabwe) and seven Western and Central African countries (Benin, Cote d'Ivoire, Ghana, Guinea, Togo, Congo Democratic Republic, and Rwanda). Country-fixed effects, region-fixed effects, year-fixed effects, standard errors clustered at the sub-national level, and sub-national population weights are used in all regressions. Significance levels: *** 1%, ** 5%, * 10%.

Looking at the subnational per-capita GNI well-being indicator, column 3 of Table 5 indicates positive impacts of tree species diversity in both tree crop plantations and forest plantations on subnational per-capita GNI for SSA as a whole: relative to zero tree crop species, the first tree crop species raises subnational per-capita GNI by 11% and two or more tree crop species raise subnational per-capita GNI by 10%,[10] both effects being significant at the 5% and 10% levels of significance, respectively; relative to zero planted forest species, the first planted forest species raises per-capita subnational per-capita GNI by 7% and two or more planted forest species raise subnational per-capita GNI by 9%,[11] both effects being

[10] The first effect size is (0.25536 / 2.362) * 100 = 10.81%. The second effect size is: (0.23142 / 2.362) * 100 = 9.80%.

[11] The first effect size is (0.17613 / 2.413) * 100 = 7.30%. The second effect size is: (0.21748 / 2.413) * 100 = 9.01%.

significant at the 5% and 1% levels of significance, respectively. A gender lens substantiates that such positive impacts on income are mainly felt by men when considering tree species diversity in tree crop plantations, while forest plantations with two or more species increase women's income by 5% and men's income by 6% relative to zero planted forest species (Table 5, columns 1 and 2).[12] A sub-regional disaggregated analysis (Table 6, columns 1-3) indicates that the positive impacts on subnational per-capita GNI associated with increased tree species diversity are only observed in Western and Central Africa countries (Panel B), but not in Eastern and Southern Africa countries (Panel A). Further, it appears that only men's income is positively affected in Western and Central Africa, but not female's income (Table 6, Panel B, columns 1-2).

Regarding the subnational HDI well-being indicator, the estimations show no overall impacts of tree species diversity (Table 5, columns 4-6). This is confirmed by the sub-regional disaggregated findings (Table 6, columns 4-6), where no impacts are observed in Western and Central African countries (Panel B), and Eastern and Southern African countries even experience negative marginal impacts on the female human development index (Panel A).

Turning to the subnational IWI well-being indicator, findings across all countries (Table 5, column 7) reveal that an increase in tree species diversity for tree crop plantations leads to an increase in the mean subnational IWI, and so does a rise in the diversity of forest plantations from zero to one tree species. Sub-regional estimations (Table 6, column 7) corroborate such impact features, not in Eastern and Southern Africa (Panel A), but only in Western and Central Africa (Panel B).

[12] The first effect size is (0.10330 / 2.068) * 100 = 4.99%. The second effect size is: (0.19109 / 3.268) * 100 = 5.85%.

Table 6: Multi–way fixed effects OLS estimation results for the impact of diversity scores of tree species on income and human development (dis-aggregated within SSA)

Panel A: Eastern and Southern Africa	(1)	(2)	(3)	(4)	(5)	(6)	(7)
VARIABLES	GNI female	GNI male	GNI	HDI female	HDI male	HDI	IWI
Diversity score (count) of tree crop species: one	0.06885	0.20653	0.05479	-0.00780*	-0.00628	-0.00481	1.20097
	(0.10724)	(0.16098)	(0.10330)	(0.00406)	(0.00397)	(0.00452)	(1.00935)
Diversity score (count) of planted forest species: one	0.01492	0.09646	0.15976	-0.00631*	-0.00431*	-0.01034**	0.00470
	(0.12018)	(0.20634)	(0.09839)	(0.00346)	(0.00245)	(0.00410)	(0.86840)
Mean age at first birth of women aged 20-50	0.18510	0.24426	0.14316	-0.00134	0.00055	-0.01525*	0.74324
	(0.17267)	(0.14582)	(0.13674)	(0.00786)	(0.00830)	(0.00890)	(1.52413)
Mean years schooling (years)	0.39437***	0.57446***	0.50209***	0.03210***	0.03184***	0.02265***	3.59033***
	(0.06288)	(0.09104)	(0.08601)	(0.00415)	(0.00421)	(0.00622)	(0.77371)
Share of seats in parliament, female (percent held by women)	0.00349	-0.00130	0.00950**	-0.00059***	-0.00030	0.00136***	-0.03716
	(0.00409)	(0.00452)	(0.00416)	(0.00018)	(0.00020)	(0.00023)	(0.05153)
Observations	458	458	1,116	458	458	1,116	1,116
R-squared	0.98387	0.98929	0.98049	0.99086	0.98825	0.96463	0.97305
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable with zero tree crop species	2.297	3.081	2.422	0.463	0.506	0.450	26.42
Mean dependent variable with zero planted forest species	1.954	2.540	1.891	0.432	0.478	0.415	26.88
Panel B: Western and Central Africa	(1)	(2)	(3)	(4)	(5)	(6)	(7)
VARIABLES	GNI female	GNI male	GNI	HDI female	HDI male	HDI	IWI
Diversity score (count) of tree crop species: one	0.10701	0.36732**	0.34302***	0.00415	0.00228	-0.00323	3.36172***
	(0.12424)	(0.16153)	(0.10025)	(0.00489)	(0.00352)	(0.00384)	(0.78613)
Diversity score (count) of tree crop species: two or more	0.08160	0.43754**	0.33074***	0.00443	0.00193	-0.00306	3.03053***
	(0.12832)	(0.17035)	(0.09578)	(0.00512)	(0.00370)	(0.00550)	(0.89229)
Diversity score (count) of planted forest species: one	0.09252	0.20175**	0.21306***	0.00441	0.00360	-0.00094	2.42033***
	(0.07037)	(0.09982)	(0.07886)	(0.00315)	(0.00256)	(0.00418)	(0.73928)
Diversity score (count) of planted forest species: two or more	0.05803	0.19027**	0.23200***	-0.00176	0.00083	0.00423	1.01939
	(0.05793)	(0.08908)	(0.08189)	(0.00516)	(0.00472)	(0.00932)	(0.97337)
Mean age at first birth of women aged 20-50	0.17122*	0.40190**	0.23881**	0.00023	-0.00698	-0.00759*	2.48602**
	(0.09693)	(0.15235)	(0.09003)	(0.00624)	(0.00545)	(0.00438)	(1.18069)
Mean years schooling (years)	-0.01305	-0.05257	0.00938	0.01592***	0.01909***	0.01622***	-1.51417*
	(0.05924)	(0.08312)	(0.06596)	(0.00328)	(0.00277)	(0.00404)	(0.88622)
Share of seats in parliament, female (percent held by women)	0.00441	-0.00832	0.00328	0.00183***	0.00191***	0.00353***	0.05860
	(0.00455)	(0.00845)	(0.00524)	(0.00043)	(0.00034)	(0.00029)	(0.05075)
Observations	485	485	1,137	485	485	1,137	1,137
R-squared	0.97481	0.97878	0.96926	0.97939	0.98180	0.96658	0.94204
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable with zero tree crop species	1.753	2.717	2.284	0.430	0.513	0.449	30.16
Mean dependent variable with zero planted forest species	2.134	3.689	2.767	0.442	0.532	0.466	32.48

Notes: This table reports the estimation results for the effects of diversity scores of tree species on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP) and the Human Development Index (HDI) by gender in sub-Saharan Africa using country-region-year fixed effects. Country-region-year observations cover twelve sub-Saharan African countries over the period of 1992 through 2016. The estimation sample includes five Eastern and Southern African countries (Kenya, Lesotho, Mozambique, Namibia, and Zimbabwe) in Panel A and seven Western and Central African countries (Benin, Cote d'Ivoire, Ghana, Guinea, Togo, Congo Democratic Republic, and Rwanda) in Panel B. Country-fixed effects, region-fixed effects, year-fixed effects, standard errors clustered at the sub-national level, and sub-national population weights are used in all regressions. Significance levels: *** 1%, ** 5%, * 10%.

4.3 What are the important factors that mediate the effects of tree planting on human well-being?

We illustrate the mediation analysis by probing the hypothesized paths for the effects of tree planting on human well-being in Western and Central Africa, where we earlier found impacts on subnational GNI and IWI (Table 3). We focus on the quadrennial treatment in this section and check consistency in the findings using the quinquennial treatment in Supplemental file (Tables S5 and S6). Although we use an appropriate method to conduct the mediation analysis, we guard against interpreting the findings as absolute causal drivers because our analysis does not go beyond the hypothesized mediating variables included in the theory of change in Figure 1. Possible mediators of the impact of tree planting on human well-being might also take a certain time to materialize and would require the occurrence of structural transformation processes to influence human well-being outcomes. It can also be emphasized that tree planting represents just one among endless development interventions that could contribute to changes in GNI and IWI.

Per-capita income

The effects of increasing the area under tree plantations on subnational income per capita (GNI) are led by employment opportunities for men outside of the agricultural sector. Column one of Table 7 presents the GNI impact regression augmented with the potential mediating variables. It shows that, after controlling for all candidates to mediation, only men's employment related variables explain GNI in the augmented impact regression. The interaction of men's agricultural employment with the dummy variable for non-negative values of PDSI is significant at the 10% critical level, indicating that when there are no drought conditions (non-negative PDSI) the higher the labor force participation for men in agriculture, the lower the income per capita; despite lower incidence of drought and favorable conditions for agriculture, shifting away from agriculture raises GNI. This is also shown by the negative sign taken by the individual effect of men's agricultural employment in column one.

Column 2 of Table 7 reveals that after removing all effects associated with men's employment in column one, the residual effect of tree planting on income becomes insignificant. This validates men's non-agricultural employment as the pathway through which increases in the area of planted trees within the last four years improve income for the local people.

Returning to the possible paths from the treatment to the GNI outcome described in Figure 1, the evidence from Table 7 suggests *Path 2*. All mediators taken together, the tree species diversification has no significant effect on GNI, and thus, rules out *Path 1*. As the interaction of the positive PDSI and the agricultural employment is a negative correlate to GNI, *Path 3* is also ruled out. Given the results in column 2 and the evidence that the percent of employed men working in agriculture has a strongly negative effect on GNI, Path 2 is retained. Moreover, as the percent of employment women working in agriculture exhibits no statistically significant relationship with GNI, tree-planting-induced income improvements were led by the non-agricultural employment opportunities accruing to men, not women.

International Wealth Index (IWI)

The effects of increasing the area under tree plantations on subnational IWI are led by enhancing income per capita. Column 3 of Table 7 first identifies several separate mediators for IWI. For example, the diversity score of tree crop species is a marginally positive correlate of IWI in the sense that the presence of two or more tree crop species does not influence IWI, while the presence of only one tree crop species reduces IWI. Planted forest species diversification is also irrelevant for IWI in the sense that its higher values are associated with lower values of IWI. PDSI is positively correlated with IWI as the decreased prevalence of droughts is associated with improvements in IWI. Employment of men and women also contribute to variations in IWI and interact strongly with GNI. When fewer women work in agriculture, as GNI rises, so does IWI; when more men work in agriculture, as GNI rises, so does IWI. Beyond employment, GNI is positively correlated with IWI.

Column 4 of Table 7 reveals that, after removing all effects associated with subnational per-capita GNI, the residual effect of tree planting on the subnational wealth index becomes insignificant and validates the GNI channel as the sole impact pathway from tree planting to the wealth index. Adding the precedent results from the first two columns of Table 7, by offering opportunities to work outside

of agriculture to men, tree-planting expansions would allow not only improvements in GNI but also improvements in wealth accumulation at the sub-national level.

Taking stock of this mediation analysis and revisiting Figure 1, *Path 2* is the retained pathway to IWI. *Paths 4 and 5* are ruled out because they don't transit through GNI. *Paths 1 and 3* are also ruled out because column 4 of Table 7 neither de-mediates environmental factors nor tree species diversification variables. This leaves out non-agricultural employment paving the way for enhanced GNI and IWI outcomes. Nevertheless, it is important to reiterate that the statistical relationships presented here only provide a starting point for understanding the factors that influence complex human well-being outcomes such as GNI and IWI.

Table 7: Mediation analysis for the impact of tree planting on income per capita and wealth index in Western and Central Africa using the quadrennial treatment

VARIABLES	(1) GNI	(2) Male Employment Demediated GNI	(3) IWI	(4) GNI Demediated IWI
Increased area under tree planting within the last 4 years (quadrennial)	0.1431 (0.2510)	0.1558 (0.1168)	6.4260** (2.5113)	0.4772 (1.5903)
Mean age at first birth of women aged 20-50	0.0591* (0.0346)	0.0623* (0.0349)	1.6212*** (0.5101)	2.3444*** (0.7634)
Mean years schooling (years)	0.2410*** (0.0208)	0.2397*** (0.0213)	1.4891*** (0.3338)	1.5681*** (0.3727)
Share of seats in parliament, female (percent held by women)	0.0053*** (0.0017)	0.0064*** (0.0018)	0.1486*** (0.0299)	0.2032*** (0.0322)
Diversity score (count) of tree crop species: one	0.0049 (0.2047)		-3.4049* (1.8804)	
Diversity score (count) of tree crop species: two or more	0.0094 (0.2148)		-3.0590 (2.1904)	
Diversity score (count) of planted forest species: one	0.0304 (0.0881)		-0.4638 (0.7270)	
Diversity score (count) of planted forest species: two or more	0.0383 (0.1160)		-3.3871*** (1.2797)	
PDSI is non-negative	0.0686 (0.0644)		1.9546** (0.7993)	
GNI per capita in thousands of USD (2011 PPP)			3.8757*** (0.6846)	
Percent of employed women working in agriculture	-0.0017 (0.0018)		0.1787*** (0.0549)	
x PDSI is non-negative	0.0010 (0.0010)		-0.0101 (0.0163)	
x GNI			-0.1200*** (0.0238)	
Percent of employed men working in agriculture	-0.0124*** (0.0024)		-0.4562*** (0.0565)	
x PDSI is non-negative	-0.0022* (0.0013)		-0.0269 (0.0194)	
x GNI			0.1900*** (0.0188)	
Observations	1,137	1,137	1,137	1,137
Number of countryadmin1code	56	56	56	56
Treatment	Quadrennial	Quadrennial	Quadrennial	Quadrennial
Method	IV	IV	IV	IV
Country-region and year fixed effects	Yes	Yes	Yes	Yes
Bootstrapped standard errors	Yes	Yes	Yes	Yes
Kleibergen-Paap F-stat	26.317	26.317	26.466	26.466
Overidentification test of all instruments: Chi-square=	3.157	3.157	2.160	2.160
Overidentification test of all instruments: p-value=	0.076	0.076	0.142	0.142
Endogeneity test of endogenous regressors: Chi-square=	0.330	0.330	9.206	9.206
Endogeneity test of endogenous regressors: p-value=	0.566	0.566	0.002	0.002

Notes: This table reports the estimation results to apprehend the mediating factors for the effects of tree planting on income per capita and wealth index at the sub-national level in Western and Central Africa over the period of 1992 through 2016. The dependent variable is the Gross National Income (GNI) per capita in thousands of USD (2011 PPP) in column one. The dependent variable in column two is the GNI per capita demeaned by all effects of male employment in column one. The dependent variable is the International Wealth Index (IWI) in column three. The dependent variable in column four is the IWI demeaned by all effects of GNI per capita in column three. IV is used due to its consistency relative to OLS in columns four and six of Table 3. Significance levels: *** 1%, ** 5%, * 10%.

5 Discussion and study limitations

We discuss our findings evidencing the human well-being impact of tree plantations and their possible drivers, focusing on what appears exceptionally relevant for decision support in the context of a huge keen interest in tree planting-based interventions for ecosystem restoration toward socio-ecological well-being. Human well-being outcomes are most often considered secondary or poorly considered in such interventions with blind surfing. Our findings offer nuanced conditional effects, highlighting the importance of developing tailored mechanisms to meet regional variations in needs, demands, and contexts.

We uncover that tree/forest plantations can have a positive impact on GNI per capita (Table 2 and Table 3) and IWI (Table 2, Table 3, and Table 4). The impacts on GNI per capita and IWI are supported by improvements in non-agricultural employment of the local people (Table 7, Tables S.5-S.6[13]). Our findings underscore the potential of tree planting as a strategy for economic growth and wealth accumulation and illustrate different routes for impacts regarding the poverty reduction potential. Because the notion of well-being is closely related to poverty and income status or level (Rojas, 2015), the well-being impacts of tree planting interventions and plantations are often used interchangeably with related impacts on poverty or income (Hajjar et al., 2021a; Hajjar et al., 2021b; Razafindratsima et al., 2021). Recent studies on poverty impacts of tree systems evidence four routes of impacts, including "1) helping people move out of poverty; 2) maintaining well-being levels through subsistence, food security, health, and cultural and spiritual values; 3) preventing declines by mitigating risks and stabilizing consumption; and 4) decreasing well-being by generating negative externalities that could cause households to fall into poverty" (Razafindratsima et al., 2021- p1). Our findings align with routes 1 and 2 and further understanding that income enhancement through non-agriculture employment is the major possible pathway for both routes. When and where positive impacts on income exist, these can primarily stem from off-farm jobs as evidenced in the literature (Adams et al., 2016; Jindal et al., 2012). They can also come from supporting economic activities of fuelwood production and NTFPs sales, which can additionally generate time savings for women and youth to engage in other/extra lucrative activities (Jones and Rosenstock, 2021). Harvesting timber and fuelwood can also reduce household expenses on such items as well as reduce time to gather fuelwood, allowing more free time to invest in additional profit-making endeavors (Razafindratsima et al. 2021).

The evidenced human effects of plantations are heterogeneous within sub-Saharan Africa. Tree/forest plantations consistently enhance GNI per capita in Western and Central Africa, where the tree plantations largely grow within a time window of three to five years, but not in other locations where the tree plantations do not grow over time as a long-term investment. This suggests that the availability of necessary resources and infrastructures for managing and monitoring tree planting and plantations can affect the realization of expected benefits and accrued well-being effects. This also entails developing or putting in place the technical expertise or necessary knowledge that may help to sustain tree planting activities and plantations via more protection, enhanced management, and monitoring. The latter elements are often less emphasized, although they are instrumental to strengthen and endure the positive impacts on income and wealth accumulation (Banin et al., 2023).

The benefits of tree plantations for improved human well-being can vary depending on the types of trees and the underlying purpose of planting. Our findings suggest that planting certain tree species has the potential to improve income and human development for both genders. Although our study could not identify the nature of these species as our descriptive analysis indicates, we assume that such skewed positive impact would materialize in locations or geographic contexts where the planted tree species appeared relevant with some level of socio-economic and cultural use-values. This would align with existing evidence on the economic significance, through increasing household income, of planted fruit trees on farms Uganda, Ethiopia, Malawi, Nigeria, and Tanzania (Miller et al., 2020; 2017). Considering the specific needs for regional livelihoods, health, economic demands, and other opportunity costs and benefits when developing tree planting interventions appear crucial to reduce the social risks of tree planting schemes and realize the call for achieving social, biodiversity, and climate-related outcomes in restoration endeavors (Di Sacco et al., 2021). More broadly, our findings imply that planting appropriate

[13] See Supplemental File, Section 1.1

and diverse tree species can contribute to improving human well-being outcomes. Regions with unidentified tree species in our dataset had higher GNI and HDI for both genders. Identifying and favoring those species in areas where they are scarce, but are valued and needed, can benefit people, particularly in ESA.

Further, our findings support the need for future inquiries into the benefits of planting different types of tree species, especially the advantages of planting oil palm versus other local species as part of ensuring an aspirational and needed level of biodiversity in tree planting initiatives. Our findings suggest that plantations of oil palm trees in Western and Central Africa have a positive correlation with GNI and HDI. While this evidence aligns with a narrow body of literature that supports the social welfare benefits from oil palm plantations (Ayompe et al., 2021; Tabe-Ojong et al., 2022; 2023; Rosdin et al., 2023), it is highly sensitive to the deep entrenched contestations and polarized debates and viewpoints on oil palm plantations. The latter most often cast oil palm plantations as a devil for their negative environmental impacts (Savilaasko et al., 2014) and social inequities against the uncertain or mitigated social welfare they may bring (see for instance Meijaard and Sheil, 2019; Chiriaco et al., 2022). Moving beyond such a status quo, we argue that continually seeking a balance through improved management and governance frameworks to reduce the harms of oil palm plantations and tree plantations in general, while enhancing their potential human well-being contributions, is paramount (Yitbarek et al., 2023).

Our study embodies the needed efforts to uncover the possible mechanisms or channels that underlie evidenced human well-being impacts for better decision support (Ferraro et al., 2014; Stern et al., 2012). The mediation channel through which tree planting initiatives can positively impact well-being is complex and comes from indirect, not straightforward causal mechanisms. That the positive impacts of tree plantations on incomes in WCA are led by men's employment opportunities outside the agricultural sector (Table 7 and Table S5) supports previous understandings that planting trees may lead to improvements in economic statuses. For instance, tree plantations can offer income-generating opportunities by exploiting certain commercial tree species, even though these opportunities can be short-term and can developed at the expense of alternative land uses and livelihood activities (Malkamäki et al., 2018). However, such a mediation mechanism to raising GNI per capita suggests a non-gender inclusive pathway. Paying attention to this gender inequity is important for making both men and women benefit equally from the income-enhancing outcome of tree/forest plantations. Continual evidence of differentiated direct and indirect mechanisms can help in better articulating how tree-planting initiatives could help achieve the assumed win-win social and environmental goals and benefits. We offer a solid conceptual framing of the possible mechanisms that could be further investigated with empirical research. More empirical research must identify the mechanisms behind the human benefits of tree planting initiatives more accurately, spanning both non-economic benefits and direct and indirect economic benefits, such as income, wealth, and other human development variables. This can further enhance evidence-based articulation of the relations between the environmental benefits and human benefits of tree/forest plantations.

Overall, the evidence offered in our study on the human well-being benefits of tree/forests plantations suggests that ensuring and maximizing such impacts could become part of pathways for establishing sustainable social benefits that would encourage the safeguarding of ecosystems. Our findings imply that the effectiveness of tree-planting interventions will not only depend on the environmental reason for planting trees but also and more importantly on ensuring that the types of trees planted equally support income and human well-being (Martin et al., 2021).

Study limitations

Despite the valuable insights provided by this study, there are five main limitations that should be acknowledged. The first limitation pertains to the tree-planting data used in the analysis. While efforts were made to use high-quality remote sensing data to provide a long-term and continental picture at a high resolution, it is only as good as the input training data (Du et al., 2022).

A second limitation is that we could not access household panel data for our analysis. The use of region-level data may not accurately reflect household-level impacts. While efforts were made to mitigate this limitation by using subnational-year instead of country-year data, future research should consider using longitudinal household-year data to further improve understanding of the impact of tree plantations on

human well-being in SSA.

A third limitation linked to using subnational-year data is that environmental variables such as CO^2 emissions and material footprint per capita are missing. This means that while explaining the effects of an environmental intervention on income and human development, we cannot make an in-depth investigation of its environmental impact pathway. We rely on the PDSI to make connections between tree planting, drought occurrence, and agricultural activities. But we are cognizant that many human-environment interactions could be displayed and explored in the mediation analysis if data is available.

A fourth limitation regards the lack of subnational-year data on primary forests spanning the twelve studied countries from 1992 to 2016 to conduct more robust analyses. Both new tree plantations and primary forests contribute to human well-being. Using the available country-year data on eleven countries from 2000 to 2016, sourced from Global Forest Watch, we present robustness checks suggesting that the estimated effects of tree planting treatment do not conflate the effects of new tree plantations and those of primary forests. Nevertheless, a clear distinction between native forests and plantations at the subnational-year level would give more comprehensive evidence if data existed.

Another limitation is that the study has focused on globally driven monetary and asset-based indicators of well-being. This is considered a narrow view of well-being, given the multidimensionality and complexity of the concept (Agarwala et al., 2014; Summers et al., 2012). Considering other dimensions of well-being, including its non-income and non-material metrics, its subjective indicators, as well as other locally and contextually relevant indicators (Loveridge et al., 2020; Woodhouse et al., 2015) would provide additional enhanced insights into the human impact potential of tree planting and plantations (Hajjar et 2021b).

Overall, although this study contributes to improving understanding of the benefits of expanding tree plantations in SSA, further research is still needed to fully apprehend their impacts on multiple dimensions of human well-being in different contexts and with more accurate and comprehensive data sources that include both environmental and household-level data.

6 Conclusion

Tree planting interventions are increasingly being considered for forest and landscape restoration initiatives, with assumptions of social co-benefits, such as improvements in the socio-economic and health conditions for people in the communities where trees are planted. The risks for policy decisions made from questionable and unclear evidence on the effectiveness of tree-planting interventions warrants an investigation to reassess empirically the impacts of tree plantations on economic and human well-being outcomes. Focusing on sub-Saharan Africa (SSA), our study conducts this reassessment by using recent longitudinal data on tree plantations (tree crops and planted forests) in the first-level administrative divisions of 12 SSA countries spanning Western, Central, Eastern, and Southern geographies in a 25-year period (1992-2016). Using increases in areas under tree planting at the extensive margin within a three-to-four-year time and species diversity in tree plantations as the treatment variables, we estimate the treatment effects on three standard well-being outcomes: Gross National Income (GNI) per capita, Human Development Index (HDI), and International Wealth Index (IWI). We add to this evidence a heterogeneity analysis across SSA geographies and genders and uncover the possible channels behind the estimated impacts.

Our investigation supports the poverty-reducing potential of tree/forest plantations as we find an expansion in areas under tree plantation over a three- to five-year period spurs a rise in sub-national income per capita by approximately 205 to 395 USD/year and IWI by more than a third. It also reveals that an increase in the number of tree crop species in the tree plantations can raise per-capita GNI by 10 to 11%, while an increase in the number of planted forest species in the tree plantations can raise per-capita GNI by 7 to 9%, in addition to positive effects of tree species diversification on IWI.

Nevertheless, the findings indicate no impact whatsoever on HDI. The empirical results also illustrate some heterogeneity in the treatment effects within the SSA geographies and across genders. The income benefits from tree planting accrue to WCA subnational regions where the tree plantations have grown as a long-term investment relatively more than the ESA subnational regions. Moreover, our analyses reveal that increasing the area under tree plantations can have positive effects on sub-national income

per capita, particularly through an improvement in off-farm employment opportunities as the major driving channel. By underscoring that the impacts on GNI per capita and IWI are led by improvements in employment outside the agricultural sector for the local people, our study supports previous understandings that planting trees may lead to improvements in economic statuses. Nonetheless, there is evidence of gender disparities in such a pathway that appears non-gender inclusive.

Our findings offer important insights into the complex relationship between tree plantations and human well-being, particularly in the context of less-covered geographies and their diverse contexts over long-time periods. Investing in sustainable tree planting interventions that ensure poverty reduction through enhanced protection, management, and monitoring can improve the well-being of people who access them. However, careful assessment of the local contexts and considering potential externalities that can generate negative impacts on human well-being is important. The findings are relevant for policymakers, development practitioners, and other stakeholders who are interested in promoting tree-planting interventions for multiple social and environmental/ecological gains. Notably, our study shows an attempt to reduce the blind surfing and uncertainties on evidence regarding the human well-being benefits of tree plantations, toward better decision-making and appropriate actions.

Methodologically, our study highlights the importance of considering sub-national data analysis to measure the effects of tree plantations in specific regions where local people live rather than measuring the effects at the national level, which may suffer from aggregation biases. It also offers solid conceptual and empirical pathways linking tree plantations to well-being benefits. However, we recognize and discuss certain caveats that further studies can consider.

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8 Appendices

8.1 Tables

Table A.1. Data Sources and Citations

Data	Time frame	Source
Tree planting	1990 to 2016	Du, Zhenrong, Le Yu, Jianyu Yang, Yidi Xu, Bin Chen, Shushi Peng, Tingting Zhang, Haohuan Fu, Nancy Harris, and Peng Gong. "A global map of planting years of plantations." *Scientific data* 9, no. 1 (2022): 141.
GNI per capita		Smits, J. & Permanyer, I. (2019). The Subnational Human Development Database. (Nature) Scientific Data, 6, 190038.
HDI, education, life expectancy, and the share of women in parliament		UNDP (United Nations Development Programme). 2022. Human Development Report 2021/2022: Uncertain Times, Unsettled Lives: Shaping our Future in a Transforming World. New York https://hdr.undp.org/data-center/documentation-and-downloads
IWI`		Smits, J. & Steendijk, R. (2015). The International Wealth Index (IWI). Social Indicators Research, 122(1), 65-85.
Population, employment, in agriculture, age at first birth of women aged 20-50, and total fertility rate		Global Data Lab available from https://globaldatalab.org/areadata)
Palmer Drought Severity Index (PDSI), from TerraClimate dataset	1960 to 2016	Abatzoglou, J.T., S.Z. Dobrowski, S.A. Parks, K.C. Hegewisch, 2018, Terraclimate, a high-resolution global dataset of monthly climate and climatic water balance from 1958-2015, Scientific Data 5:170191, doi:10.1038/sdata.2017.191 CLOSEIMPORT
Primary forest in 2000	2000, 2013, and 2016	Turubanova S., Potapov P., Tyukavina, A., and Hansen M. (2018) Ongoing primary forest loss in Brazil, Democratic Republic of the Congo, and Indonesia. Environmental Research Letters https://doi.org/10.1088/1748-9326/aacd1c
Primary forest cover extent in 2000 and annual loss in primary forest tree cover extent in 2001-2016. • Tree cover extent in 2000 in primary forest in the tropics. (Used as base) • Tree cover loss in primary forest 2001-2022. (Annual drop)	2000 to 2022	Global Forest Watch available from https://www.globalforestwatch.org/dashboards/global/
Africa subnational shapefile	Accessed by 2023	GDAM dataset
Oil palm suitability	Accessed by 2023	Strona, Giovanni, Simon D. Stringer, Ghislain Vieilledent, Zoltan Szantoi, John Garcia-Ulloa, and Serge A. Wich. "Small room for compromise between oil palm cultivation and primate conservation in Africa." *Proceedings of the National Academy of Sciences* 115, no. 35 (2018): 8811-8816.

Table A.2: First stage of the multi–way fixed effects IV estimations of the impact of tree planting in sub-Saharan Africa (triennial and quadrennial treatments)

Panel A: First stage for IV estimations using triennial treatments (years 1992-2016)	
VARIABLES	(1) Increased area under tree planting within the last 3 years
Previous share of other regions with increased tree planting area (triennial)	0.91633***
	(0.31894)
x Mean palm oil suitability index	-7.49633
	(5.02737)
Mean age at first birth of women aged 20-50	0.09183**
	(0.03938)
Mean years schooling (years)	-0.00911
	(0.02823)
Share of seats in parliament, female (percent held by women)	-0.00295
	(0.00233)
Observations	2,253
Country-region and year fixed effects	Yes
Standard errors clustered at country-region level	Yes
Population weights	Yes
Kleibergen-Paap F-stat	11.069
Panel B: First stage for IV estimations using the quadrennial treatment (years 1992-2016)	
VARIABLES	(1) Increased area under tree planting within the last 4 years
Previous share of other regions with increased tree planting area (quadrennial)	1.25760***
	(0.37195)
x Mean palm oil suitability index	-12.28861**
	(5.81191)
Mean age at first birth of women aged 20-50	0.06588*
	(0.03433)
Mean years schooling (years)	-0.01422
	(0.02247)
Share of seats in parliament, female (percent held by women)	-0.00364*
	(0.00214)
Observations	2,253
Country-region and year fixed effects	Yes
Standard errors clustered at country-region level	Yes
Population weights	Yes
Kleibergen-Paap F-stat	15.239

Notes: This table reports the first-stage regression results for the IV estimation of the effects of tree planting on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP), and the Human Development Index (HDI) in sub-Saharan Africa using country-region-year fixed effects. The dependent variable is the triennial treatment in Panel A and the quadrennial treatment in Panel B. Country-region-year observations cover twelve sub-Saharan African countries over the period of 1992 through 2016. The estimation sample includes five Eastern and Southern African countries (Kenya, Lesotho, Mozambique, Namibia, and Zimbabwe) and seven Western and Central African countries (Benin, Cote d'Ivoire, Ghana, Guinea, Togo, Congo Democratic Republic, and Rwanda). Country-fixed effects, region-fixed effects, year-fixed effects, standard errors clustered at the sub-national level, and sub-national population weights are used. Significance levels: *** 1%, ** 5%, * 10%.

Table A.3: First stage of the multi–way fixed effects IV estimations of the impact of tree planting in Western and Central Africa (triennial and quadrennial treatments)

Panel A: First stage for IV estimations using triennial treatments (years 1992-2016)	
	(1)
VARIABLES	Increased area under tree planting within the last 3 years
Previous share of other regions with increased tree planting area (triennial)	1.63408***
	(0.26194)
x Mean palm oil suitability index	-20.76019***
	(3.54695)
Mean age at first birth of women aged 20-50	0.04418
	(0.03709)
Mean years schooling (years)	-0.02497
	(0.03195)
Share of seats in parliament, female (percent held by women)	-0.00118
	(0.00263)
Observations	1,137
Country-region and year fixed effects	Yes
Standard errors clustered at country-region level	Yes
Population weights	Yes
Kleibergen-Paap F-stat	19.816
Panel B: First stage for IV estimations using the quadrennial treatment (years 1992-2016)	
	(1)
VARIABLES	Increased area under tree planting within the last 4 years
Previous share of other regions with increased tree planting area (quadrennial)	2.01026***
	(0.28791)
x Mean palm oil suitability index	-26.45605***
	(4.11164)
Mean age at first birth of women aged 20-50	0.03090
	(0.02851)
Mean years schooling (years)	-0.02607
	(0.01975)
Share of seats in parliament, female (percent held by women)	-0.00201
	(0.00190)
Observations	1,137
Country-region and year fixed effects	Yes
Standard errors clustered at country-region level	Yes
Population weights	Yes
Kleibergen-Paap F-stat	24.402

Notes: This table reports the first-stage regression results for the IV estimation of the effects of tree planting on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP) and the Human Development Index (HDI) in Western and Central Africa using country-region-year fixed effects. The dependent variable is the triennial treatment in Panel A and the quadrennial treatment in Panel B. Country-region-year observations cover the period of 1992 through 2016. The estimation sample includes seven Western and Central African countries (Benin, Cote d'Ivoire, Ghana, Guinea, Togo, Congo Democratic Republic, and Rwanda). Country-fixed effects, region-fixed effects, year-fixed effects, standard errors clustered at the sub-national level, and sub-national population weights are used. Significance levels: *** 1%, ** 5%, * 10%.

Table A.4: First stage of the multi–way fixed effects IV estimations of the impact of tree planting in Eastern and Southern Africa (triennial and quadrennial treatments)

Panel A: First stage for IV estimations using triennial treatments (years 1992-2016)	
	(1)
VARIABLES	Increased area under tree planting within the last 3 years
Previous share of other regions with increased tree planting area (triennial)	-19.50091***
	(6.08293)
x Mean palm oil suitability index	270.75521***
	(76.28874)
Mean age at first birth of women aged 20-50	0.11011
	(0.07175)
Mean years schooling (years)	-0.05842
	(0.05439)
Share of seats in parliament, female (percent held by women)	-0.00261
	(0.00236)
Observations	1,116
Country-region and year fixed effects	Yes
Standard errors clustered at country-region level	Yes
Population weights	Yes
Kleibergen-Paap F-stat	21.967
Panel B: First stage for IV estimations using the quadrennial treatment (years 1992-2016)	
	(1)
VARIABLES	Increased area under tree planting within the last 4 years
Previous share of other regions with increased tree planting area (quadrennial)	-19.94358***
	(6.25150)
x Mean palm oil suitability index	275.26748***
	(78.39692)
Mean age at first birth of women aged 20-50	0.07477
	(0.06715)
Mean years schooling (years)	-0.06783
	(0.05899)
Share of seats in parliament, female (percent held by women)	-0.00306*
	(0.00163)
Observations	1,116
Country-region and year fixed effects	Yes
Standard errors clustered at country-region level	Yes
Population weights	Yes
Kleibergen-Paap F-stat	15.573

Notes: This table reports the first-stage regression results for the IV estimation of the effects of tree planting on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP) and the Human Development Index (HDI) in Eastern and Southern Africa using country-region-year fixed effects. The dependent variable is the triennial treatment in Panel A and the quadrennial treatment in Panel B. Country-region-year observations cover the period of 1992 through 2016. The estimation sample includes five Eastern and Southern African countries (Kenya, Lesotho, Mozambique, Namibia, and Zimbabwe). Country-fixed effects, region-fixed effects, year-fixed effects, standard errors clustered at the sub-national level, and sub-national population weights are used. Significance levels: *** 1%, ** 5%, * 10%.

Table A.5: Interdependence of the spatial instrument, mean outcomes of the neighboring subnational regions, and the outcomes

Panel A: OLS regressions of the mean outcomes of the neighboring subnational regions on the spatial instrument

	(1) Mean GNI of neighbors	(2) Mean HDI of neighbors	(3) Mean IWI of neighbors	(4) Mean GNI of neighbors	(5) Mean HDI of neighbors	(6) Mean IWI of neighbors
VARIABLES						
Previous share of other regions with increased tree planting area (triennial)	-0.03780	0.04564	2.83726			
	(0.16434)	(0.03053)	(2.34957)			
x Mean palm oil suitability index	-0.35349	-0.54953	20.51070			
	(2.46991)	(0.47684)	(36.39713)			
Previous share of other regions with increased tree planting area (quadrennial)				-0.11780	0.04556	0.58345
				(0.17342)	(0.03210)	(2.86839)
x Mean palm oil suitability index				-0.07639	-0.56602	47.81895
				(2.56609)	(0.50105)	(42.82434)
Observations	2,700	2,700	2,252	2,700	2,700	2,252
R-squared	0.98243	0.95910	0.95707	0.98254	0.95892	0.95608
Tree planting treatment	Triennial	Triennial	Triennial	Quadrennial	Quadrennial	Quadrennial
Method	OLS	OLS	OLS	OLS	OLS	OLS
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable	2.079	0.426	28.07	2.079	0.426	28.07

Panel B: OLS regressions of the outcomes on the mean outcomes of the neighboring subnational regions

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Mean GNI per capita in thousands of USD (2011 PPP) of neighbors	0.86731***			0.86731***		
	(0.14534)			(0.14534)		
Mean Human Development Index (value) of neighbors		0.26850			0.26850	
		(0.16343)			(0.16343)	
Mean International Wealth Index (IWI) of neighbors			1.04109***			1.04109***
			(0.21070)			(0.21070)
Observations	2,700	2,700	2,252	2,700	2,700	2,252
R-squared	0.96875	0.94314	0.95930	0.96875	0.94314	0.95930
Tree planting treatment	Triennial	Triennial	Triennial	Quadrennial	Quadrennial	Quadrennial
Method	OLS	OLS	OLS	OLS	OLS	OLS
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable	2.235	0.445	28.05	2.235	0.445	28.05

Notes: This table reports the interdependence of the spatial instrument and the mean outcomes of the neighboring subnational regions (Panel A) and the interdependence of the mean outcomes of the neighboring subnational regions and the outcomes (Panel B). GNI: Gross National Income per capita in thousands of USD (2011 PPP). HDI: Human Development Index (value); IWI: International Wealth Index (IWI). A neighboring subnational region to a subnational region of interest is defined as being a subnational region of a country bordering the country of the subnational region of interest or as being another subnational region of the country of the subnational region of interest. When the country of the subnational region of interest has no bordering country in the study sample (Kenya, Lesotho, and Namibia), we use as neighbors all subnational regions of the sample other than the subnational region of interest. Significance levels: *** 1%, ** 5%, * 10%.

8.2 Figures

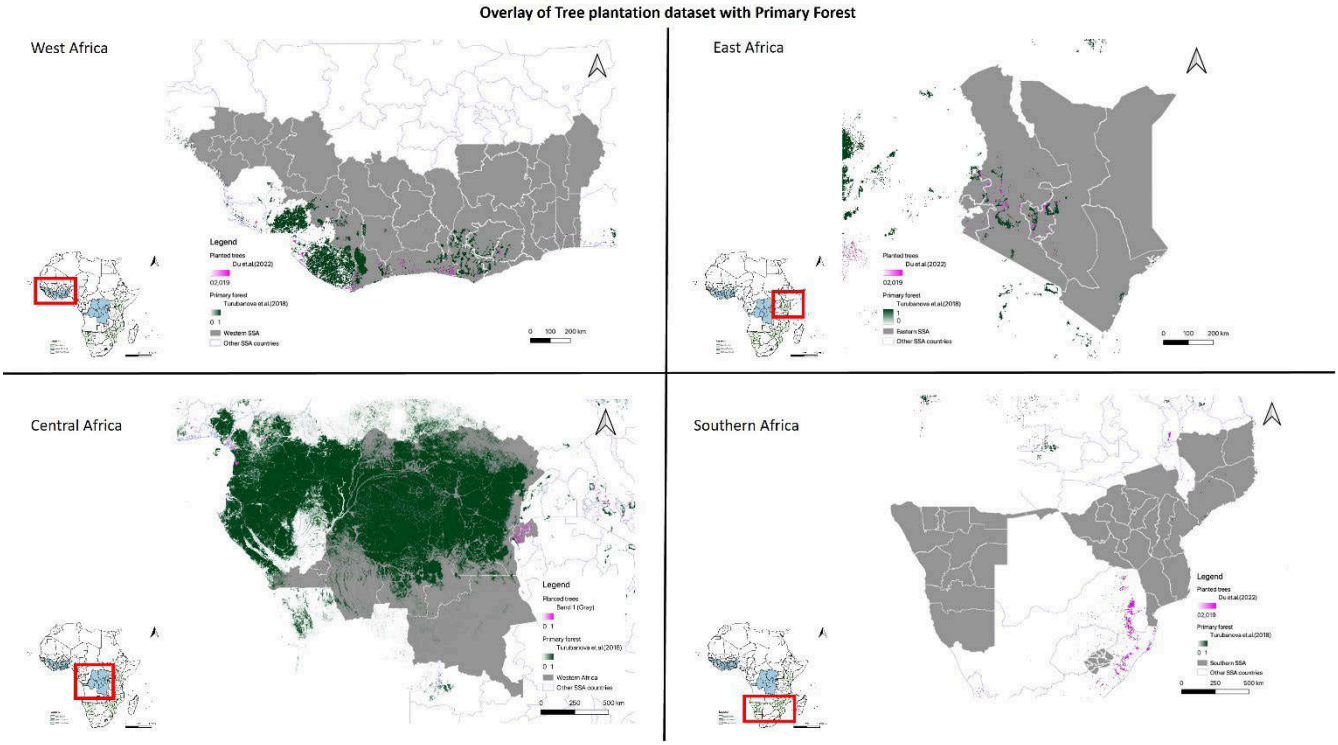


Figure A1: Map of tree plantation dataset (Due et al., 2022) overlaid with map of primary forest (Turubanova et al., 2018)

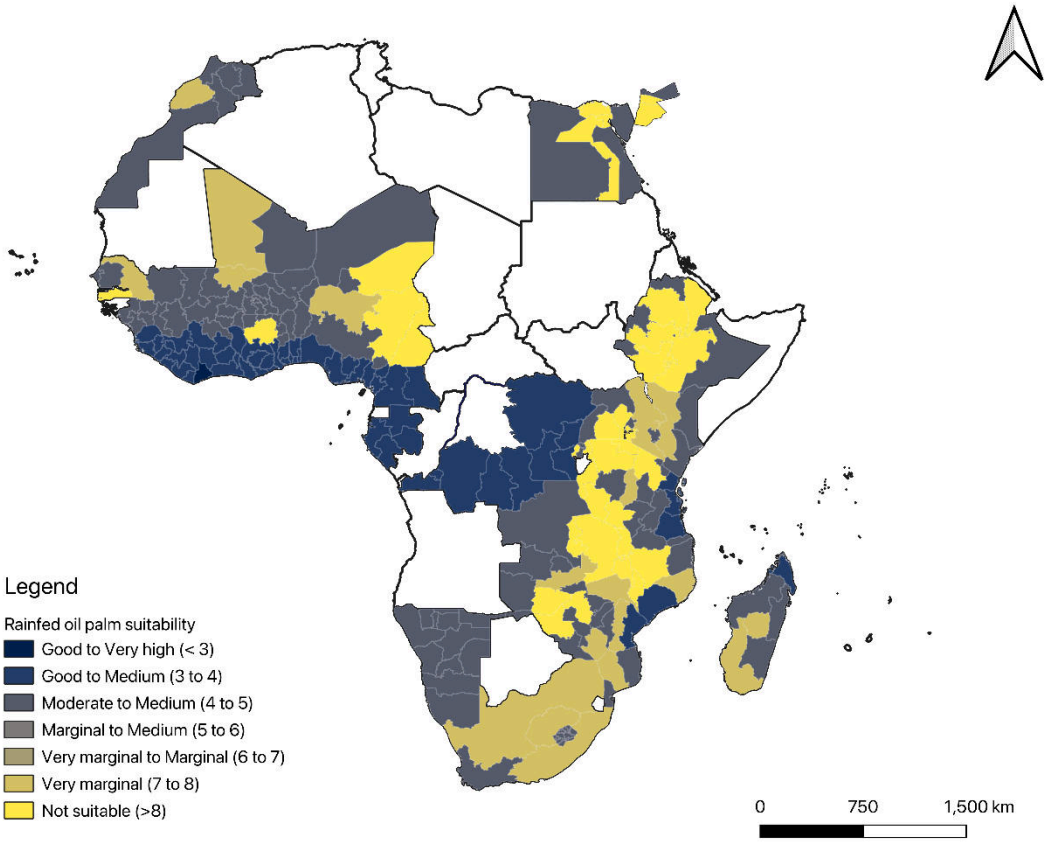


Figure A2. Mean oil palm suitability at country sub-national level.

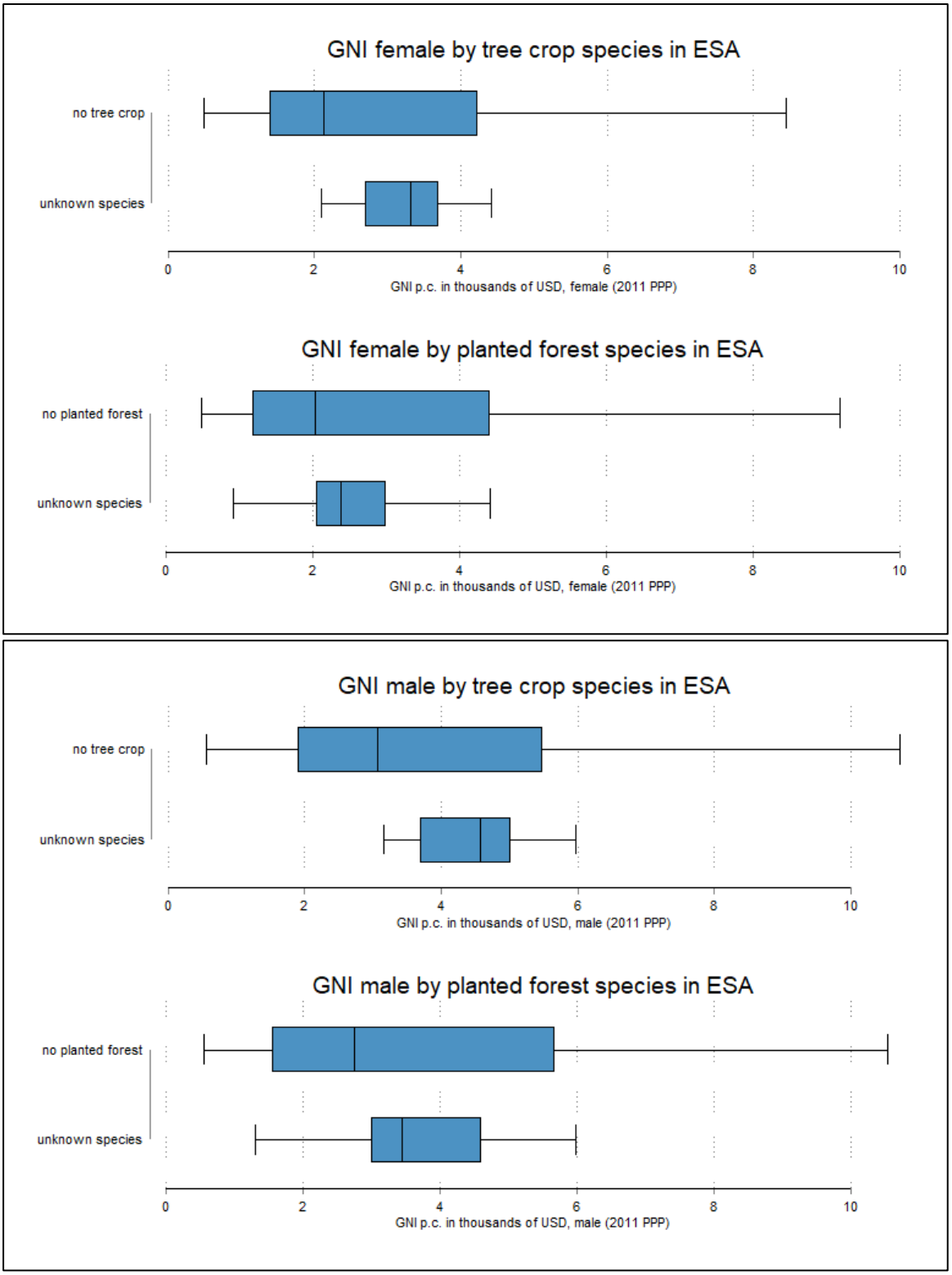


Figure A.3: Distribution of subnational-year GNI female and male by tree species in the Eastern and Southern Africa sub-sample

Notes: Outside values of box plots are excluded. ESA: Eastern and Southern Africa. Countries: Kenya, Lesotho, Mozambique, Namibia, and Zimbabwe. Years: 2000-2016. The distribution of diversity of tree crop species in the country-region-year sub-sample is: no tree crop (95%), unknown species (5%). The distribution of diversity of planted forest species in the country-region-year sub-sample is: no planted forest (75%), unknown species (25%).

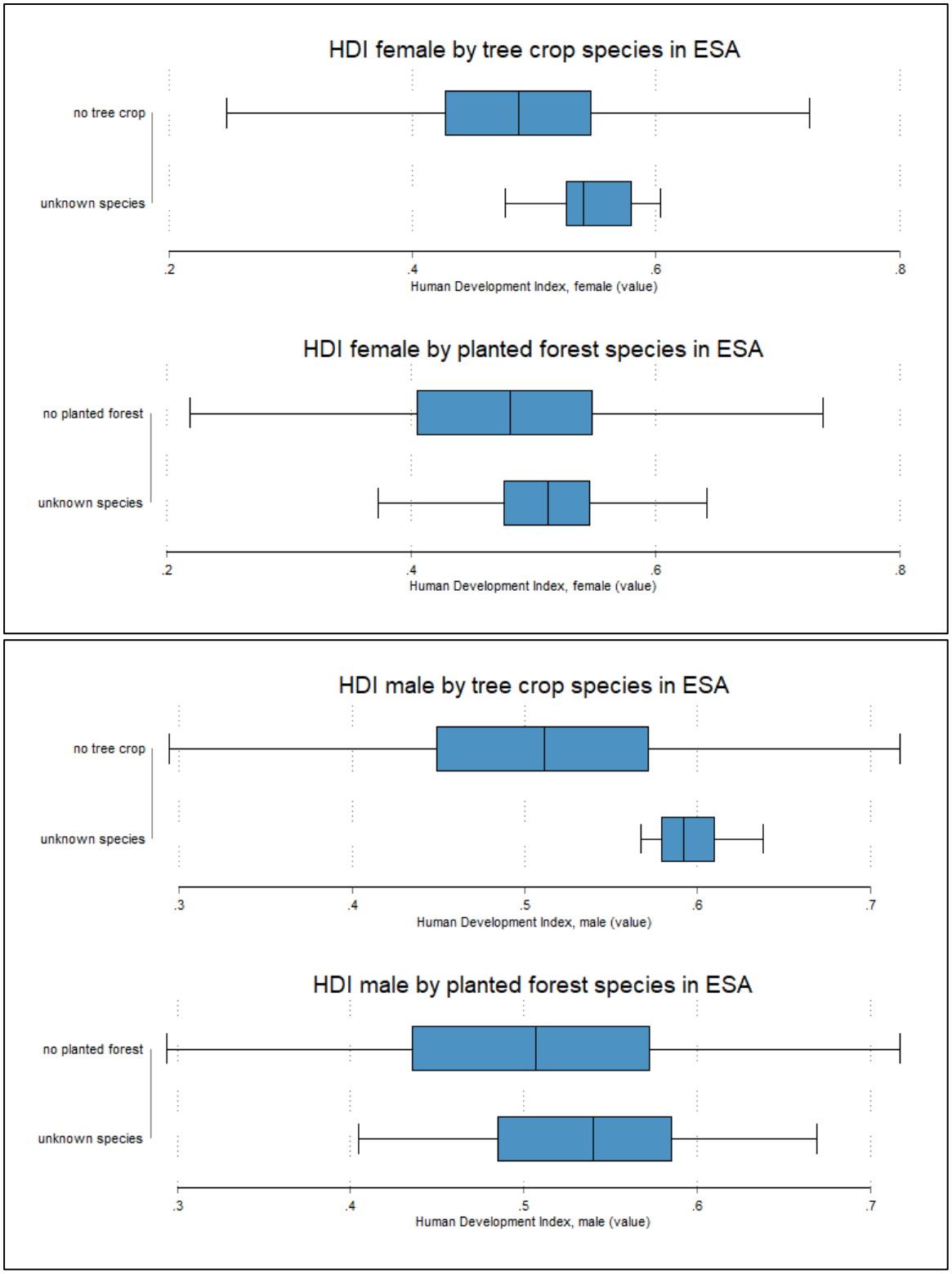


Figure A.4: Distribution of subnational-year HDI female and male by tree species in the Eastern and Southern Africa sub-sample

Notes: Outside values of box plots are excluded. ESA: Eastern and Southern Africa. Countries: Kenya, Lesotho, Mozambique, Namibia, and Zimbabwe. Years: 2000-2016. The distribution of diversity of tree crop species in the country-region-year sub-sample is: no tree crop (95%), unknown species (5%). The distribution of diversity of planted forest species in the country-region-year sub-sample is: no planted forest (75%), unknown species (25%).

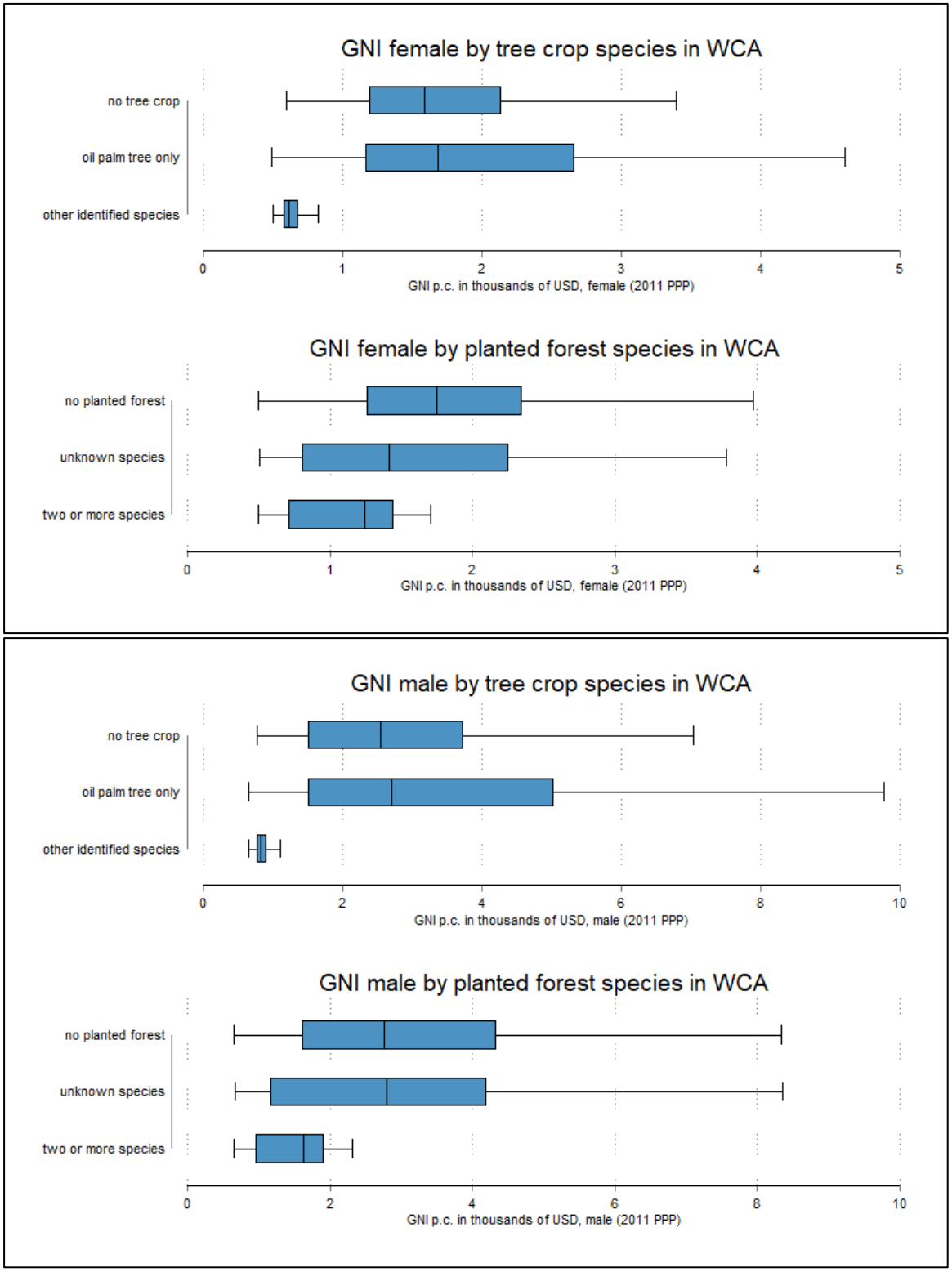
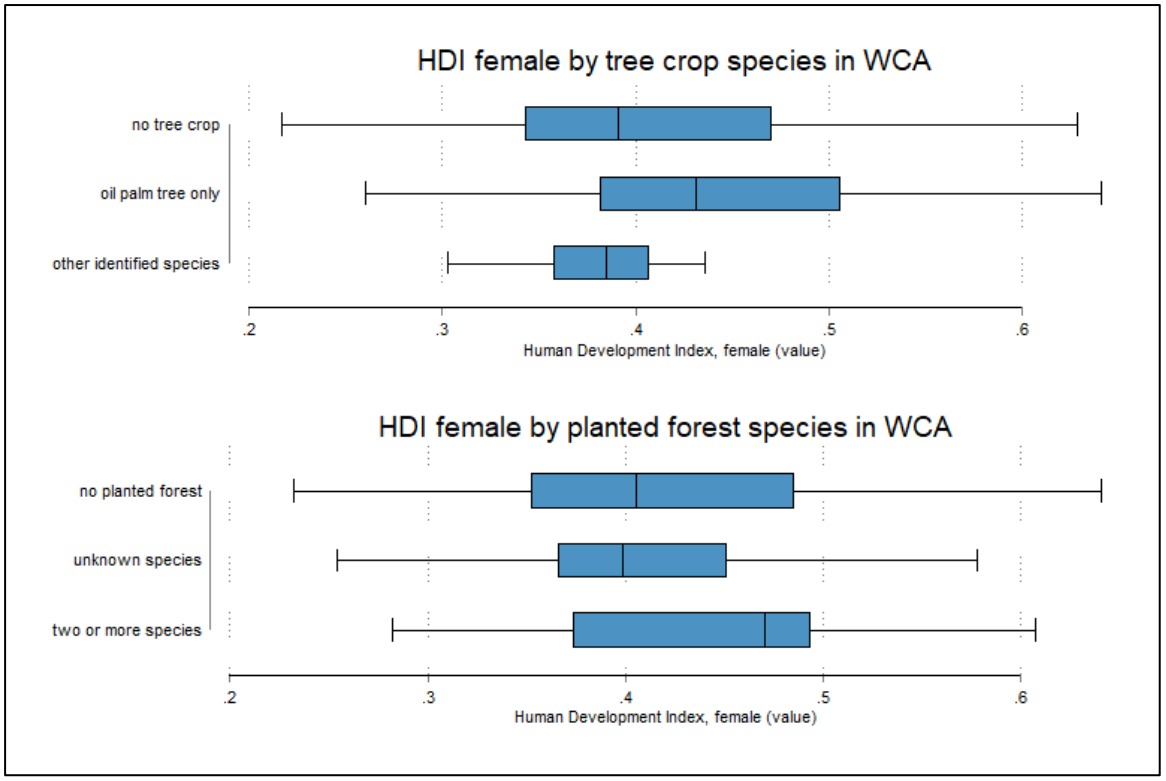


Figure A.5: Distribution of subnational-year GNI female and male by tree species in the Western and Central Africa sub-sample

Notes: Outside values of box plots are excluded. WCA: Western and Central Africa. Countries: Benin, Congo Democratic Republic, Cote d'Ivoire, Ghana, Guinea, Rwanda, and Togo. Years: 2000-2016. The distribution of diversity of tree crop species in the country-region-year sub-sample is: no tree crop (52.0%), oil palm tree only (43%), other identified species (with or without oil palm trees) (5%). The last category includes: palm trees only; oil palm and rubber trees; oil palm and rubber trees; oil palm and coconut trees; oil palm, rubber, and coconut trees.The distribution of diversity of planted forest species in the country-region-year sub-sample is: no planted forest (63%), unknown species (24%); and two or more species (13%). The last category includes: East Asian white birch and unknown species; Bocote and unknown species; East Asian white birch, pine, and unknown species; East Asian white birch, eucalyptus, and unknown species; East Asian white birch, pine, cypress, and unknown species; East Asian white birch, eucalyptus, cypress, and unknown species; East Asian white birch, eucalyptus, pine, and unknown species; East Asian white birch, eucalyptus, cypress, pine, and unknown species.



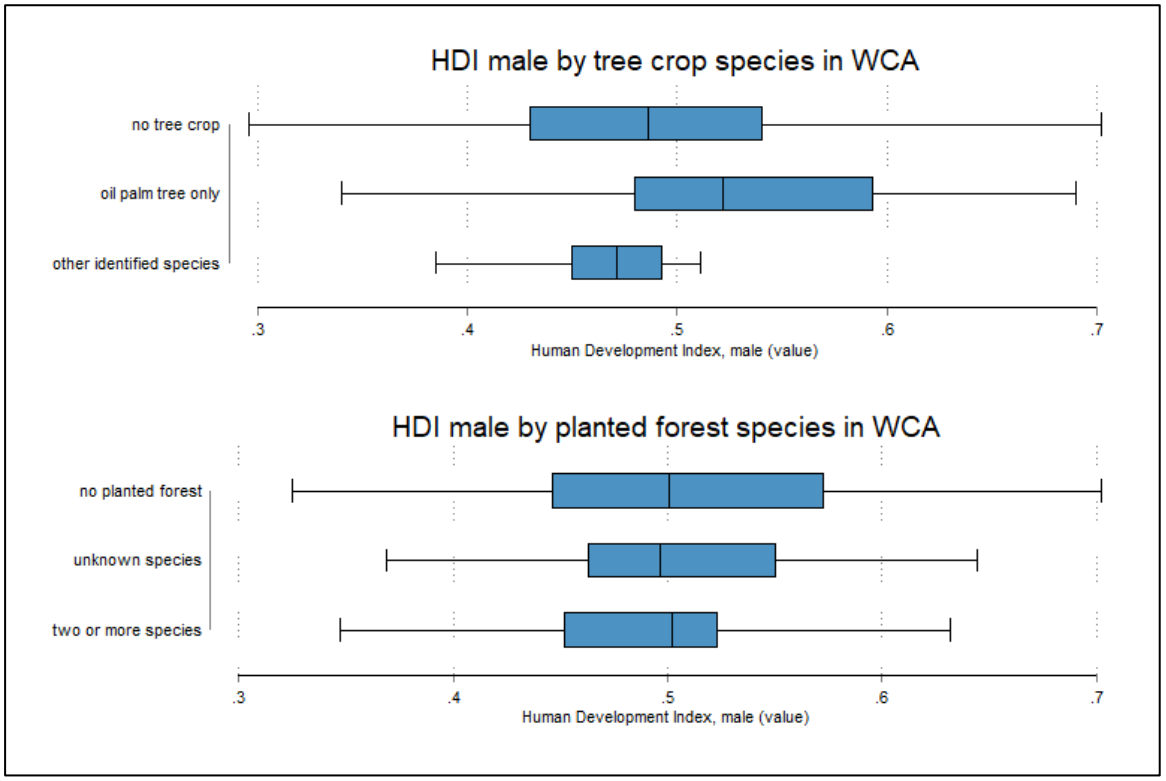


Figure A.6: Distribution of subnational-year HDI female and male by tree species in the Western and Central Africa sub-sample

Notes: Outside values of box plots are excluded. WCA: Western and Central Africa. Countries: Benin, Congo Democratic Republic, Cote d'Ivoire, Ghana, Guinea, Rwanda, and Togo. Years: 2000-2016. The distribution of diversity of tree crop species in the country-region-year sub-sample is: no tree crop (52.0%), oil palm tree only (43%), other identified species (with or without oil palm trees) (5%). The last category includes: palm trees only; oil palm and rubber trees; oil palm and rubber trees; oil palm and coconut trees; oil palm, rubber, and coconut trees. The distribution of diversity of planted forest species in the country-region-year sub-sample is: no planted forest (63%), unknown species (24%); and two or more species (13%). The last category includes: East Asian white birch and unknown species; Bocote and unknown species; East Asian white birch, pine, and unknown species; East Asian white birch, eucalyptus, and unknown species; East Asian white birch, pine, cypress, and unknown species; East Asian white birch, eucalyptus, cypress, and unknown species; East Asian white birch, eucalyptus, pine, and unknown species; East Asian white birch, eucalyptus, cypress, pine, and unknown species.

9 Supplemental Materials: Robustness checks

- **Using a quinquennial treatment**

Allowing for tree planting to influence well-being within five years, we find consistent and insignificant OLS estimates for HDI (Table S1, Panel A, column 2) and consistent and significant IV estimates for GNI and IWI (Table S1, Panel B, columns 1 and 3). The positive effects of the quinquennial treatment on GNI and IWI are of similar magnitudes to those obtained using the quadrennial tree planting treatment (Table 2, Panel B, columns 4 and 6): an increase in IWI by 40% and an increase in income per capita by 205 USD per year.[14]

Separate estimations for Western and Central Africa as well as Eastern and Southern Africa (Table S3) reveal that tree planting effects on the Gross National Income (GNI) per capita are observed only in Western and Central Africa (Panel B, columns 1 and 4), similarly to the findings based on the triennial and quadrennial treatments (Tables 3 and 4, Panel B, columns 1 and 4). In addition, tree planting does not enhance HDI within a five-year time window, but its effects on IWI are generally positive, with a lower magnitude and a marginal significance in Eastern and Southern Africa.

Mediation analysis of the impacts of the quinquennial treatment on GNI (Table S5) and IWI (Table S6) in WCA yields similar conclusions to those drawn using the quadrennial treatment in Table 7. Regarding GNI, men's non-agricultural employment remains the mediator (Table S5 and Path 2 to GNI in Figure 6). Regarding IWI, results based on IV estimations in the first two columns of Table S6 indicate the earlier finding that IWI is GNI-mediated, and employment-led income enhancement improves IWI. Due to the overidentification test results in column 1 of Table S6, we also conducted the mediation analysis using OLS estimations, which indicate multiple mediators, such as GNI, employment, and diversification.

[14] The effect size is (10.90685 / 27.11) * 100 = 40.23% for IWI and (1.02582 / 2.134) * 100 = 48.07% for GNI using the quinquennial treatment.

Table S1: Multi–way fixed effects OLS and IV estimation results for the impact of tree planting on income and human development in sub-Saharan Africa (using a quinquennial treatment)

Panel A: OLS estimations			
	(1)	(2)	(3)
VARIABLES	GNI	HDI	IWI
Increased area under tree planting within the last 5 years (quinquennial)	0.16244**	-0.00353	1.29581**
	(0.07181)	(0.00342)	(0.59338)
Mean age at first birth of women aged 20-50	0.20236**	-0.00663	1.97696*
	(0.07763)	(0.00430)	(1.02369)
Mean years schooling (years)	0.09360*	0.01528***	-0.68829
	(0.05429)	(0.00294)	(0.71467)
Share of seats in parliament, female (percent held by women)	0.00608	0.00285***	0.07203
	(0.00523)	(0.00043)	(0.05184)
Observations	2,253	2,253	2,253
R-squared	0.97304	0.95787	0.95012
Tree planting treatment	Quinquennial	Quinquennial	Quinquennial
Country-region and year fixed effects	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes
Population weights	Yes	Yes	Yes
Mean dependent variable in the absence of treatment	2.134	0.424	27.11
Testing tree planting effect = 0, p-value =	0.0257	0.305	0.0312
Panel B: IV estimations			
	(1)	(2)	(3)
VARIABLES	GNI	HDI	IWI
Increased area under tree planting within the last 5 years (quinquennial)	1.02582**	0.06601	10.90685**
	(0.42696)	(0.04091)	(4.37498)
Mean age at first birth of women aged 20-50	0.15525*	-0.01103**	1.53358
	(0.08308)	(0.00546)	(1.05980)
Mean years schooling (years)	0.10881*	0.01650***	-0.50652
	(0.05727)	(0.00350)	(0.72973)
Share of seats in parliament, female (percent held by women)	0.01136**	0.00327***	0.12911***
	(0.00461)	(0.00061)	(0.04863)
Observations	2,253	2,253	2,253
Country-region and year fixed effects	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes
Population weights	Yes	Yes	Yes
Kleibergen-Paap F-stat	9.759	9.759	9.759
Overidentification test of all instruments: Chi-square=	0.716	4.872	0.337
Overidentification test of all instruments: p-value=	0.398	0.027	0.562
Endogeneity test of endogenous regressors: Chi-square=	3.524	0.568	4.593
Endogeneity test of endogenous regressors: p-value=	0.060	0.451	0.032

Notes: This table reports the estimation results for the effects of tree planting on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP), and the Human Development Index (HDI) in sub-Saharan Africa using country-region-year fixed effects. Country-region-year observations cover twelve sub-Saharan African countries over the period of 1992 through 2016. The estimation sample includes five Eastern and Southern African countries (Kenya, Lesotho, Mozambique, Namibia, and Zimbabwe) and seven Western and Central African countries (Benin, Cote d'Ivoire, Ghana, Guinea, Togo, Congo Democratic Republic, and Rwanda). Significance levels: *** 1%, ** 5%, * 10%.

Table S.2: First stage of the multi–way fixed effects IV estimations of the impact of tree planting in sub-Saharan Africa (quinquennial treatment)

VARIABLES	(1) Increased area under tree planting within the last 5 years
Previous share of other regions with increased tree planting area (triennial)	0.86959**
	(0.40466)
x Mean palm oil suitability index	-7.09889
	(6.43994)
Mean age at first birth of women aged 20-50	0.04131
	(0.03071)
Mean years schooling (years)	-0.01101
	(0.02017)
Share of seats in parliament, female (percent held by women)	-0.00445**
	(0.00195
Observations	2,253
Country-region and year fixed effects	Yes
Standard errors clustered at country-region level	Yes
Population weights	Yes
Kleibergen-Paap F-stat	9.759

Notes: This table reports the first-stage regression results for the IV estimation of the effects of tree planting on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP), and the Human Development Index (HDI) in sub- Saharan Africa using country-region-year fixed effects. The dependent variable is the quinquennial treatment variable. Country-region-year observations cover twelve sub-Saharan African countries over the period of 1992 through 2016. The estimation sample includes five Eastern and Southern African countries (Kenya, Lesotho, Mozambique, Namibia, and Zimbabwe) and seven Western and Central African countries (Benin, Cote d'Ivoire, Ghana, Guinea, Togo, Congo Democratic Republic, and Rwanda). Country-fixed effects, region-fixed effects, year-fixed effects, standard errors clustered at the sub-national level, and sub-national population weights are used. Significance levels: *** 1%, ** 5%,* 10%.

Table S3: Multi–way fixed effects OLS and IV estimation results for the impact of tree planting on income and human development by subdivision of sub-Saharan Africa (using a quinquennial treatment)

Panel A: OLS estimations	(1)	(2)	(3)	(4)	(5)	(6)
	WCA	WCA	WCA	ESA	ESA	ESA
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Increased area under tree planting within the last 5 years (quinquennial)	0.27725***	-0.00413	2.11591***	0.06033	-0.00147	0.97774
	(0.07808)	(0.00290)	(0.70164)	(0.09488)	(0.00736)	(0.65859)
Mean age at first birth of women aged 20-50	0.22595**	-0.00728	2.32861*	0.13600	-0.01335	0.77794
	(0.08701)	(0.00449)	(1.18124)	(0.13558)	(0.00851)	(1.51129)
Mean years schooling (years)	0.00048	0.01624***	-1.58487*	0.50452***	0.02128***	3.65718***
	(0.06685)	(0.00413)	(0.88149)	(0.09214)	(0.00651)	(0.80484)
Share of seats in parliament, female (percent held by women)	0.00315	0.00353***	0.05749	0.00925**	0.00143***	-0.03271
	(0.00526)	(0.00029)	(0.04989)	(0.00407)	(0.00024)	(0.05195)
Observations	1,137	1,137	1,137	1,116	1,116	1,116
R-squared	0.96922	0.96654	0.94533	0.98046	0.96392	0.97309
Tree planting treatment	Quinquennial	Quinquennial	Quinquennial	Quinquennial	Quinquennial	Quinquennial
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable in the absence of treatment	2.289	0.432	28.90	2.028	0.419	25.89
Testing tree planting effect = 0, p-value =	0.000796	0.160	0.00388	0.528	0.842	0.144
Panel B: IV estimations	(1)	(2)	(3)	(4)	(5)	(6)
	WCA	WCA	WCA	ESA	ESA	ESA
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Increased area under tree planting within the last 5 years (quinquennial)	0.75251**	-0.07203**	12.02957***	0.17784	0.00591	6.61004*
	(0.34665)	(0.02734)	(3.44164)	(0.28765)	(0.02860)	(3.37435)
Mean age at first birth of women aged 20-50	0.21116**	-0.00485	2.01444	0.11565	-0.01629	0.32786
	(0.08791)	(0.00417)	(1.22989)	(0.13927)	(0.00980)	(1.47925)
Mean years schooling (years)	-0.00220	0.01659***	-1.65007*	0.51208***	0.02195***	4.21968***
	(0.06734)	(0.00450)	(0.90180)	(0.10199)	(0.00744)	(0.99344)
Share of seats in parliament, female (percent held by women)	0.00691	0.00299***	0.13365**	0.00988**	0.00144***	-0.00950
	(0.00521)	(0.00028)	(0.05273)	(0.00436)	(0.00026)	(0.06313)
Observations	1,137	1,137	1,137	1,116	1,116	1,116
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap F-stat	28.008	28.008	28.008	9.082	9.082	9.082
Overidentification test of all instruments: Chi-square=	0.000	4.024	0.341	3.257	2.749	2.681
Overidentification test of all instruments: p-value=	0.999	0.045	0.559	0.071	0.097	0.102
Endogeneity test of endogenous regressors: Chi-square=	3.528	1.407	7.314	0.391	0.547	5.779
Endogeneity test of endogenous regressors: p-value=	0.060	0.236	0.007	0.532	0.460	0.016

Notes: This table reports the estimation results for the effects of tree planting on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP), and the Human Development Index (HDI) in sub-Saharan Africa using country-region-year fixed effects. Country-region-year observations cover sub-Saharan African countries over the period of 1992 through 2016. The estimation sample includes seven Western and Central African countries (Benin, Cote d'Ivoire, Ghana, Guinea, Togo, Congo Democratic Republic, and Rwanda) in the first three columns and five Eastern and Southern African countries (Kenya, Lesotho, Mozambique, Namibia, and Zimbabwe) in the last three columns. Significance levels: *** 1%, ** 5%, * 10%.

Table S.4: First stage of the multi–way fixed effects IV estimations of the impact of tree planting by subdivision of sub-Saharan Africa (quinquennial treatment)

Panel A: First stage for IV estimations in Western and Central Africa (years 1992-2016)	
VARIABLES	(1) Increased area under tree planting within the last 5 years
Previous share of other regions with increased tree planting area (triennial)	1.79550***
	(0.30831)
x Mean palm oil suitability index	-20.96342***
	(5.26634)
Mean age at first birth of women aged 20-50	0.01654
	(0.02427)
Mean years schooling (years)	-0.01284
	(0.01502)
Share of seats in parliament, female (percent held by women)	-0.00303*
	(0.00177)
Observations	1,137
Country-region and year fixed effects	Yes
Standard errors clustered at country-region level	Yes
Population weights	Yes
Kleibergen-Paap F-stat	28.008
Panel B: First stage for IV estimations in Eastern and Southern Africa (years 1992-2016)	
VARIABLES	(1) Increased area under tree planting within the last 5 years
Previous share of other regions with increased tree planting area (quadrennial)	-21.15692***
	(7.99699)
x Mean palm oil suitability index	287.52904***
	(100.36690)
Mean age at first birth of women aged 20-50	0.04271
	(0.06846)
Mean years schooling (years)	-0.08276
	(0.06700)
Share of seats in parliament, female (percent held by women)	-0.00344
	(0.00217)
Observations	1,116
Country-region and year fixed effects	Yes
Standard errors clustered at country-region level	Yes
Population weights	Yes
Kleibergen-Paap F-stat	9.082

Notes: This table reports the first-stage regression results for the IV estimation of the effects of tree planting on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP), and the Human Development Index (HDI) in sub- Saharan Africa using country-region-year fixed effects. The dependent variable is the quinquennial treatment variable. Country-region-year observations cover twelve sub-Saharan African countries over the period of 1992 through 2016. The estimation sample includes seven Western and Central African countries (Benin, Cote d'Ivoire, Ghana, Guinea, Togo, Congo Democratic Republic, and Rwanda) in Panel A and five Eastern and Southern African countries (Kenya, Lesotho, Mozambique, Namibia, and Zimbabwe) in Panel B. Country-fixed effects, region- fixed effects, year-fixed effects, standard errors clustered at the sub-national level, and sub-national population weights are used. Significance levels: *** 1%, ** 5%, * 10%.

Table S5: Mediation analysis for the impact of tree planting on income per capita in Western and Central Africa using the quinquennial treatment

VARIABLES	(1) GNI	(2) Male Employment Demediated GNI	(3) GNI	(4) Male Employment Demediated GNI
Increased area under tree planting within the last 5 years (quinquennial)	-0.0080	0.0926	-0.0062	0.0606
	(0.2709)	(0.1421)	(0.0457)	(0.0430)
Mean age at first birth of women aged 20-50	0.0653*	0.0706**	0.0653*	0.0738**
	(0.0348)	(0.0351)	(0.0334)	(0.0339)
Mean years schooling (years)	0.2414***	0.2460***	0.2414***	0.2492***
	(0.0205)	(0.0220)	(0.0206)	(0.0204)
Share of seats in parliament, female (percent held by women)	0.0053***	0.0061***	0.0053***	0.0060***
	(0.0017)	(0.0019)	(0.0017)	(0.0018)
Diversity score (count) of tree crop species: one	0.1197		0.1185*	
	(0.1980)		(0.0682)	
Diversity score (count) of tree crop species: two or more	0.1266		0.1254	
	(0.1973)		(0.0838)	
Diversity score (count) of planted forest species: one	0.0697		0.0691	
	(0.1071)		(0.0598)	
Diversity score (count) of planted forest species: two or more	0.0761		0.0754	
	(0.1391)		(0.0936)	
PDSI is non-negative	0.0657		0.0658	
	(0.0645)		(0.0638)	
Percent of employed women working in agriculture	-0.0020		-0.0020	
	(0.0018)		(0.0017)	
x PDSI is non-negative	0.0011		0.0011	
	(0.0010)		(0.0010)	
Percent of employed men working in agriculture	-0.0132***		-0.0131***	
	(0.0024)		(0.0020)	
x PDSI is non-negative	-0.0022*		-0.0022*	
	(0.0013)		(0.0013)	
Observations	1,137	1,137	1,137	1,137
R-squared			0.3647	0.2737
Number of countryadmin1code	56	56	56	56
Treatment	Quinquennial	Quinquennial	Quinquennial	Quinquennial
Method	IV	IV	OLS	OLS
Country-region and year fixed effects	Yes	Yes	Yes	Yes
Bootstrapped standard errors	Yes	Yes	Yes	Yes
Kleibergen-Paap F-stat	24.593	24.593		
Overidentification test of all instruments: Chi-square=	2.902	2.902		
Overidentification test of all instruments: p-value=	0.088	0.088		
Endogeneity test of endogenous regressors: Chi-square=	0.000	0.000		
Endogeneity test of endogenous regressors: p-value=	0.993	0.993		

Notes: This table reports the estimation results to apprehend the mediating factors for the effects of tree planting on income per capita at the sub-national level in Western and Central Africa over the period of 1992 through 2016. The estimation method is IV in the first two columns and OLS in the last two columns. The dependent variable is the Gross National Income (GNI) per capita in thousands of USD (2011 PPP) in columns one and three. The dependent variable in column two is the GNI per capita demeaned by all effects of male employment in column one. The dependent variable in column four is the GNI per capita demeaned by all effects of male employment in column three. Significance levels: *** 1%, ** 5%, * 10%.

Table S6: Mediation analysis for the impact of tree planting on wealth index in Western and Central Africa using the quinquennial treatment

VARIABLES	(1) IWI	(2) GNI Demediated IWI	(3) IWI	(4) GNI Demediated IWI	(5) Diversification and GNI Demediated IWI	(6) Diversification Employment and GNI Demediated IWI
Increased area under tree planting within the last 5 years (quinquennial)	9.4503***	1.2755	0.2501	1.5181***	0.9784*	0.3114
	(3.2768)	(2.0574)	(0.4729)	(0.4747)	(0.5414)	(0.4883)
Mean age at first birth of women aged 20-50	1.6309***	2.3657***	1.7958***	2.1166***	2.0633***	1.8456***
	(0.5363)	(0.7900)	(0.4505)	(0.6797)	(0.6833)	(0.4548)
Mean years schooling (years)	1.3603***	1.4573***	1.4892***	1.4091***	1.3481***	1.3743***
	(0.3531)	(0.3966)	(0.3142)	(0.3505)	(0.3536)	(0.3124)
Share of seats in parliament, female (percent held by women)	0.1503***	0.2040***	0.1554***	0.2093***	0.2205***	0.1674***
	(0.0304)	(0.0331)	(0.0279)	(0.0315)	(0.0318)	(0.0288)
Diversity score (count) of tree crop species: one	-5.0922**		1.2447**			
	(2.2433)		(0.5932)			
Diversity score (count) of tree crop species: two or more	-4.4058*		1.7269			
	(2.4559)		(1.2139)			
Diversity score (count) of planted forest species: one	-1.9436*		1.0306**			
	(1.1245)		(0.4120)			
Diversity score (count) of planted forest species: two or more	-5.2657***		-1.9873**			
	(1.7265)		(0.8597)			
PDSI is non-negative	1.9427**		1.8632***			
GNI per capita in thousands of USD (2011 PPP)	(0.8174)		(0.7221)			
	3.4966***		4.4943***			
	(0.7472)		(0.5442)			
Percent of employed women working in agriculture	0.1863***		0.1619***			
	(0.0564)		(0.0538)			
x PDSI is non-negative	-0.0100		-0.0054			
	(0.0175)		(0.0147)			
x GNI	-0.1226***		-0.1207***			
	(0.0241)		(0.0233)			
Percent of employed men working in agriculture	-0.4803***		-0.4527***			
	(0.0575)		(0.0539)			
x PDSI is non-negative	-0.0278		-0.0287			
	(0.0201)		(0.0177)			
x GNI	0.2050***		0.1778***			
	(0.0206)		(0.0178)			
Observations	1,137	1,137	1,137	1,137	1,137	1,137
R-squared			0.6823	0.2279	0.2209	0.2304
Number of countryadmin1code	56	56	56	56	56	56
Treatment	Quinquennial	Quinquennial	Quinquennial	Quinquennial	Quinquennial	Quinquennial
Method	IV	IV	OLS	OLS	OLS	OLS
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Bootstrapped standard errors	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap F-stat	24.526	24.526				
Overidentification test of all instruments: Chi-square=	5.802	5.802				
Overidentification test of all instruments: p-value=	0.016	0.016				
Endogeneity test of endogenous regressors: Chi-square=	14.416	14.416				
Endogeneity test of endogenous regressors: p-value=	0.000	0.000				

Notes: This table reports the estimation results to apprehend the mediating factors for the effects of tree planting on wealth index at the sub-national level in Western and Central Africa over the period of 1992 through 2016. The dependent variable is the International Wealth Index (IWI) in columns one through three. The dependent variable in column two is the IWI demeaned by all effects of GNI per capita in column one. The dependent variable in column four is the IWI demeaned by all effects of GNI per capita in column three. The dependent variable in column five is the IWI demeaned by all effects of GNI per capita and tree species diversification in column three. The dependent variable in column six is the IWI demeaned by all effects of GNI per capita, tree species diversification, and employment in column three. Significance levels: *** 1%, ** 5%, * 10%

- **Controlling for primary forest characteristics**

As our tree planting treatment variable regards new trees planted beyond primary forest dynamics, one may argue that the estimated human well-being effects of tree planting treatment conflate the effects of primary forests and new tree plantations. To address this concern, we re-estimate the main regressions using the logarithm of primary forest tree cover extent as a possible confounder to the effect of triennial or quadrennial treatment on subnational GNI, HDI, IWI. Two caveats apply to these re-estimations. First, subnational data on primary forest tree cover extent are available only for the years 2000, 2013, and 2016 (Turubanova et al., 2018). Second, national data on primary forest tree cover extent are available for the years 2000 through 2016 for all countries of the sample except Namibia.[15]

Discarding our subnational-year sub-sample for all 12 countries in the years 1992 through 1999 and for Namibia in all years and including the national-year logarithm of primary forest tree cover extent as an additional control in the vector $X_{ij,t}$ in Eq.2, we find that our main results using the pooled sample of WCA and ESA countries are maintained statistically significant. Comparing Panel B of Table S7 and Table 2, the new estimate of the effect of tree planting on GNI nearly doubles the previous estimate, and the new estimate of the effect of tree planting on IWI is approximately 28 to 53% larger than the previous estimate. Further heterogeneity analysis indicates that these effects on GNI and IWI are concentrated in WCA (Table S8, Panel B) and that the effect on IWI in ESA is imprecisely estimated (Table S9). Therefore, there is robust evidence that the core IV estimations in Table 2, Table 3, and Table 4, establishing our pooled and heterogeneous effects of tree planting on human well-being outcomes, even place a lower bound on the true effects, as compared to the IV estimations in Table S7, Table S8, and Table S9, that controlled for primary forest characteristics.

[15] Primary forest tree cover extent at the country level is sourced from Global Forest Watch: https://www.globalforestwatch.org/dashboards/global/. The data portal gives access to primary forest tree cover extent in hectares in the year 2000 (as a base) and the annual loss in primary forest tree cover extent in hectares since the year 2001. We then recover primary forest tree cover extent in hectares for the years 2001 through 2016 by subtracting the cumulative loss in primary forest tree cover extent from the base primary forest tree cover extent.

Table S.7: Multi–way fixed effects OLS and IV estimation results for the impact of tree planting on income and human development in sub-Saharan Africa

Panel A: OLS estimations	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Increased area under tree planting within the last 3 years (triennial)	0.14407***	0.00140	0.90951			
	(0.03946)	(0.00179)	(0.60222)			
Increased area under tree planting within the last 4 years (quadrennial)				0.15000***	0.00058	1.10817**
				(0.04695)	(0.00192)	(0.42687)
Mean age at first birth of women aged 20-50	0.29232***	-0.00266	2.65296*	0.29506***	-0.00262	2.66854*
	(0.08939)	(0.00439)	(1.35928)	(0.09022)	(0.00438)	(1.35480)
Mean years schooling (years)	0.04768	0.01825***	-1.83604**	0.04670	0.01820***	-1.83535**
	(0.05962)	(0.00246)	(0.84110)	(0.06042)	(0.00248)	(0.85406)
Share of seats in parliament, female (percent held by women)	0.00626	0.00145***	0.15030**	0.00646	0.00145***	0.15188**
	(0.00606)	(0.00022)	(0.06541)	(0.00610)	(0.00022)	(0.06553)
Log primary forest tree cover extent in ha	-0.41488	0.06986*	-16.66120**	-0.42655	0.06887*	-16.57245**
	(0.90978)	(0.03926)	(8.14925)	(0.91919)	(0.03929)	(8.27071)
Observations	1,545	1,545	1,545	1,545	1,545	1,545
R-squared	0.97809	0.98113	0.95738	0.97794	0.98112	0.95735
Tree planting treatment	Triennial	Triennial	Triennial	Quadrennial	Quadrennial	Quadrennial
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable in the absence of treatment	1.892	0.433	28.25	1.851	0.427	28.20
Testing tree planting effect = 0, p-value =	0.000429	0.436	0.134	0.00190	0.765	0.0109
Panel B: IV estimations	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Increased area under tree planting within the last 3 years (triennial)	2.04339**	0.02150	15.26848**			
	(0.87164)	(0.01662)	(7.51023)			
Increased area under tree planting within the last 4 years (quadrennial)				1.76015**	-0.00828	15.25879*
				(0.77437)	(0.01215)	(7.86549)
Mean age at first birth of women aged 20-50	0.23565**	-0.00326	2.22451	0.27827***	-0.00253	2.52103*
	(0.09476)	(0.00457)	(1.47317)	(0.08289)	(0.00434)	(1.33608)
Mean years schooling (years)	0.14486**	0.01928***	-1.10135	0.11531*	0.01783***	-1.23235
	(0.06607)	(0.00244)	(0.79333)	(0.05875)	(0.00248)	(0.80348)
Share of seats in parliament, female (percent held by women)	0.00730	0.00146***	0.15819**	0.00948	0.00144***	0.17837**
	(0.00657)	(0.00022)	(0.07331)	(0.00592)	(0.00023)	(0.06938)
Log primary forest tree cover extent in ha	1.73105	0.09258*	-0.43787	1.19549	0.05995	-2.31742
	(1.57930)	(0.04748)	(12.01726)	(1.57977)	(0.03877)	(13.50963)
Observations	1,545	1,545	1,545	1,545	1,545	1,545
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap F-stat	3.110	3.110	3.110	3.262	3.262	3.262
Overidentification test of all instruments: Chi-square=	0.416	5.723	0.864	1.838	4.496	2.290
Overidentification test of all instruments: p-value=	0.519	0.017	0.352	0.175	0.034	0.130
Endogeneity test of endogenous regressors: Chi-square=	17.846	2.102	12.771	7.853	0.183	9.151
Endogeneity test of endogenous regressors: p-value=	0.000	0.147	0.000	0.005	0.669	0.002

Notes: This table reports the estimation results for the effects of tree planting on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP), and the Human Development Index (HDI) in sub-Saharan Africa using country-region-year fixed effects. Country-region-year observations cover eleven sub-Saharan African countries over the period of 2000 through 2016. The estimation sample includes four Eastern and Southern African countries (Kenya, Lesotho, Mozambique, and Zimbabwe) and seven Western and Central African countries (Benin, Cote d'Ivoire, Ghana, Guinea, Togo, Congo Democratic Republic, and Rwanda). Significance levels: *** 1%, ** 5%, * 10%.

Table S.8: Multi–way fixed effects OLS and IV estimation results for the impact of tree planting on income and human development in Western and Central Africa

Panel A: OLS estimations	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Increased area under tree planting within the last 3 years (triennial)	0.15307***	0.00244	1.25484*			
	(0.03887)	(0.00200)	(0.71500)			
Increased area under tree planting within the last 4 years (quadrennial)				0.14454***	0.00110	1.34827***
				(0.04808)	(0.00213)	(0.50167)
Mean age at first birth of women aged 20-50	0.30563***	-0.00351	2.96312*	0.30899***	-0.00342	2.98646*
	(0.09665)	(0.00456)	(1.50238)	(0.09809)	(0.00455)	(1.49617)
Mean years schooling (years)	-0.05243	0.01677***	-2.44560**	-0.05642	0.01665***	-2.47096**
	(0.07637)	(0.00262)	(1.06366)	(0.07771)	(0.00265)	(1.08594)
Share of seats in parliament, female (percent held by women)	-0.00215	0.00178***	0.17556**	-0.00154	0.00179***	0.18072**
	(0.00617)	(0.00030)	(0.06886)	(0.00612)	(0.00030)	(0.06878)
Log primary forest tree cover extent in ha	-0.04076	0.07728**	-13.88142	-0.07950	0.07548**	-14.03805
	(1.00744)	(0.03582)	(8.54889)	(1.01558)	(0.03613)	(8.69452)
Observations	892	892	892	892	892	892
R-squared	0.97847	0.98218	0.95440	0.97819	0.98212	0.95426
Tree planting treatment	Triennial	Triennial	Triennial	Quadrennial	Quadrennial	Quadrennial
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable in the absence of treatment	2.173	0.443	29.34	2.195	0.439	29.71
Testing tree planting effect = 0, p-value =	0.000234	0.227	0.0848	0.00398	0.608	0.00950
Panel B: IV estimations	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Increased area under tree planting within the last 3 years (triennial)	1.92012**	0.02288	16.49995**			
	(0.84432)	(0.01763)	(7.99333)			
Increased area under tree planting within the last 4 years (quadrennial)				1.62440**	-0.00192	16.17080**
				(0.64720)	(0.01390)	(7.77017)
Mean age at first birth of women aged 20-50	0.22395**	-0.00445	2.25847	0.27094***	-0.00334	2.60538*
	(0.10368)	(0.00473)	(1.68798)	(0.08740)	(0.00445)	(1.45955)
Mean years schooling (years)	0.06878	0.01817***	-1.39993	0.01022	0.01652***	-1.80352*
	(0.07929)	(0.00250)	(1.01596)	(0.06670)	(0.00273)	(0.99069)
Share of seats in parliament, female (percent held by women)	-0.00800	0.00172***	0.12513	-0.00043	0.00179***	0.19184**
	(0.00841)	(0.00028)	(0.08945)	(0.00611)	(0.00030)	(0.07290)
Log primary forest tree cover extent in ha	2.05026	0.10148**	4.15868	1.37840	0.07250*	0.56452
	(1.59084)	(0.04446)	(13.13145)	(1.54482)	(0.03700)	(14.17203)
Observations	892	892	892	892	892	892
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap F-stat	3.249	3.249	3.249	4.461	4.461	4.461
Overidentification test of all instruments: Chi-square=	0.954	7.286	0.829	3.292	5.499	1.831
Overidentification test of all instruments: p-value=	0.329	0.007	0.363	0.070	0.019	0.176
Endogeneity test of endogenous regressors: Chi-square=	13.173	1.932	12.486	6.092	0.033	9.883
Endogeneity test of endogenous regressors: p-value=	0.000	0.165	0.000	0.014	0.855	0.002

Notes: This table reports the estimation results for the effects of tree planting on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP), and the Human Development Index (HDI) in Western and Central Africa using country-region-year fixed effects. Country-region-year observations cover seven Western and Central African countries (Benin, Cote d'Ivoire, Ghana, Guinea, Togo, Congo Democratic Republic, and Rwanda) over the period of 2000 through 2016. Significance levels: *** 1%, ** 5%, * 10%.

Table S.9: Multi–way fixed effects OLS and IV estimation results for the impact of tree planting on income and human development in Eastern and Southern Africa

Panel A: OLS estimations	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Increased area under tree planting within the last 3 years (triennial)	0.14073*	0.00095	0.01236			
	(0.08024)	(0.00280)	(0.51580)			
Increased area under tree planting within the last 4 years (quadrennial)				0.21902	-0.00099	0.44657
				(0.14211)	(0.00344)	(0.51759)
Mean age at first birth of women aged 20-50	0.20825*	-0.00201	0.87636	0.21362*	-0.00205	0.88963
	(0.11743)	(0.00613)	(1.94531)	(0.11599)	(0.00610)	(1.94253)
Mean years schooling (years)	0.59037***	0.04034***	2.33006**	0.59663***	0.04019***	2.36335**
	(0.08032)	(0.00353)	(0.93441)	(0.08064)	(0.00359)	(0.94878)
Share of seats in parliament, female (percent held by women)	-0.00065	-0.00048***	-0.03415	-0.00027	-0.00047***	-0.03413
	(0.00375)	(0.00018)	(0.06602)	(0.00365)	(0.00017)	(0.06602)
Log primary forest tree cover extent in ha	-1.62346	-0.53362***	-62.01452*	-1.10142	-0.53789***	-60.61897*
	(3.27864)	(0.18496)	(30.65401)	(3.30767)	(0.18501)	(31.03629)
Observations	653	653	653	653	653	653
R-squared	0.98489	0.98825	0.97598	0.98492	0.98825	0.97599
Tree planting treatment	Triennial	Triennial	Triennial	Quadrennial	Quadrennial	Quadrennial
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable in the absence of treatment	1.606	0.422	27.14	1.546	0.417	26.87
Testing tree planting effect = 0, p-value =	0.0875	0.736	0.981	0.132	0.776	0.394
Panel B: IV estimations	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	GNI	HDI	IWI	GNI	HDI	IWI
Increased area under tree planting within the last 3 years (triennial)	0.42501	0.02718	22.62721			
	(0.34699)	(0.03247)	(14.23812)			
Increased area under tree planting within the last 4 years (quadrennial)				-0.26567	0.00556	27.19437
				(0.71306)	(0.04509)	(25.02668)
Mean age at first birth of women aged 20-50	0.21066*	-0.00179	1.06811	0.19909	-0.00185	1.69110
	(0.11655)	(0.00626)	(2.11285)	(0.13040)	(0.00582)	(2.36935)
Mean years schooling (years)	0.61163***	0.04230***	4.02149**	0.55949***	0.04070***	4.41252*
	(0.08803)	(0.00490)	(1.88923)	(0.12373)	(0.00594)	(2.59267)
Share of seats in parliament, female (percent held by women)	-0.00142	-0.00055***	-0.09593	-0.00026	-0.00047***	-0.03461
	(0.00371)	(0.00019)	(0.06208)	(0.00382)	(0.00017)	(0.05861)
Log primary forest tree cover extent in ha	-1.28065	-0.50199**	-34.74334	-2.63226	-0.51723**	23.86132
	(3.32672)	(0.20767)	(63.79746)	(2.97514)	(0.20667)	(51.41043)
Observations	653	653	653	653	653	653
Country-region and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors clustered at country-region level	Yes	Yes	Yes	Yes	Yes	Yes
Population weights	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap F-stat	1.412	1.412	1.412	0.744	0.744	0.744
Overidentification test of all instruments: Chi-square=	1.043	2.464	1.795	3.029	1.832	0.044
Overidentification test of all instruments: p-value=	0.307	0.117	0.180	0.082	0.176	0.834
Endogeneity test of endogenous regressors: Chi-square=	2.117	0.164	10.731	0.001	0.092	10.080
Endogeneity test of endogenous regressors: p-value=	0.146	0.685	0.001	0.969	0.762	0.001

Notes: This table reports the estimation results for the effects of tree planting on the International Wealth Index (IWI), the Gross National Income (GNI) per capita in thousands of USD (2011 PPP), and the Human Development Index (HDI) in Eastern and Southern Africa using country-region-year fixed effects. Country-region-year observations cover four Eastern and Southern African countries (Kenya, Lesotho, Mozambique, and Zimbabwe) over the period of 2000 through 2016. Significance levels: *** 1%, ** 5%, * 10%.